MATH4515 Statistical and Computational Methods in Financial Mathematics

Lecture Note 1

Basic Statistical Methods for Model Construction

Introduction – Prediction on "future uncertainty"

Many financial problems (such as finding the price of a financial derivative, determining optimal investment strategies in portfolio management etc.) involve prediction on the future price movement of stock price. For example:

- In determining the price of a financial derivative (e.g. options), the current price of the derivative can be found by computing the present value of the future payoff of the derivative and the calculation requires the probability distribution of the underlying asset (e.g. stock, index).
- In analyzing the optimal portfolio under mean variance portfolio framework, we need to estimate the expected returns of stocks and the correlations between the stock returns (proxied by covariance matrix)
- In estimating the Value-at-Risk (VaR) or Expected Shortfall (ES) in risk management problem, one may need to determine the probability distribution of the portfolio/stock.

In this "introductory" chapter, we shall review some basic statistical tools that are essential for data analysis involved in various financial problems (portfolio analysis, risk management etc.).

Model construction using parametric model

We let X be a random variable that we are interested in (e.g. daily stock return). One can construct a model (called parametric model) for X through the following procedure:

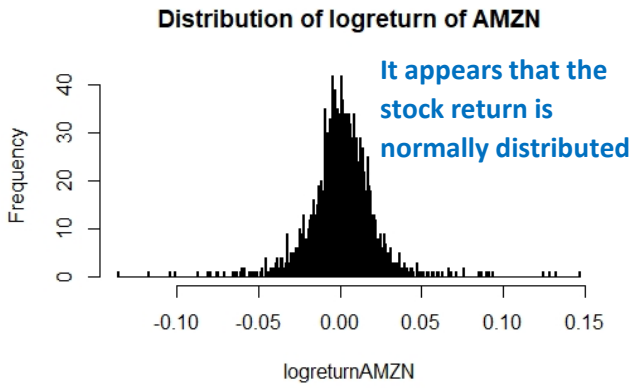
Step 1: Collect sample data for X, denoted by $\{X_1, X_2, \dots, X_n\}$.

Step 2: By examining the distribution of the sample data (through histogram), we make an assumption on the distribution of X.

Step 3: Determine the parameter(s) in the distribution using some methods.

As an example, suppose that we would like to get a model for daily return of Amazon traded in US (Stock code: AMZN)

	A	B	C	D	E	F	G	H	I
1	Date	Open	High	Low	Close	Adj Close	Volume	Daily return	Log daily return
2	3/1/11	181.369995	186	181.210007	184.220001	184.220001	5331400		
3	4/1/11	186.149994	187.699997	183.779999	185.009995	185.009995	5031800	0.004288318	0.00427915
4	5/1/11	184.100006	187.449997	184.070007	187.419998	187.419998	3418800	0.013026339	0.012942226
5	6/1/11	186.5	187.410004	185.25	185.860001	185.860001	3179700	-0.008323535	-0.00835837
6	7/1/11	187.880005	188.449997	183.740005	185.490005	185.490005	5221700	-0.001990724	-0.001992708
7	10/1/11	185.039993	185.289993	182.509995	184.679993	184.679993	3375900	-0.004366877	-0.004376439
8	11/1/11	185.419998	186	183.210007	184.339996	184.339996	2814200	-0.001841006	-0.001842703
9	12/1/11	185.360001	185.380005	183.300003	184.080002	184.080002	2679100	-0.001410405	-0.0014114
10	13/1/11	183.600006	186.449997	183.509995	185.529999	185.529999	3367000	0.007876994	0.007846132
11	14/1/11	185.5	188.940002	184.919998	188.75	188.75	3662800	0.017355689	0.017206799
12	18/1/11	188.660004	191.600006	188.25	191.25	191.25	3896200	0.013245033	0.013158085
13	19/1/11	190.899994	191	186.210007	186.869995	186.869995	3882400	-0.022901987	-0.023168312
14	20/1/11	185.289993	186.850006	181	181.960007	181.960007	5709900	-0.026274887	-0.02662624
15	21/1/11	183	183.25	176.839996	177.419998	177.419998	6806200	-0.024950587	-0.025267129
16	24/1/11	177.949997	178.490005	174.149994	176.850006	176.850006	5600300	-0.003212671	-0.003217842
17	25/1/11	175.5	176.75	174.279999	176.699997	176.699997	4651000	-0.000848227	-0.000848587
18	26/1/11	177.509995	177.889999	174.630005	175.389999	175.389999	3762200	-0.007413684	-0.007441302
19	27/1/11	177.479996	185	177.309998	184.449997	184.449997	14594400	0.051656298	0.050366348
20	28/1/11	171.449997	173.710007	166.899994	171.139999	171.139999	19908200	-0.072160467	-0.074896479
21	31/1/11	170.160004	171.440002	167.410004	169.639999	169.639999	6718100	-0.008764754	-0.00880339



$$X \sim Normal(\mu, \sigma^2)$$

$$\Downarrow$$

$$\mu \approx \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$$\sigma^2 = \cdots$$

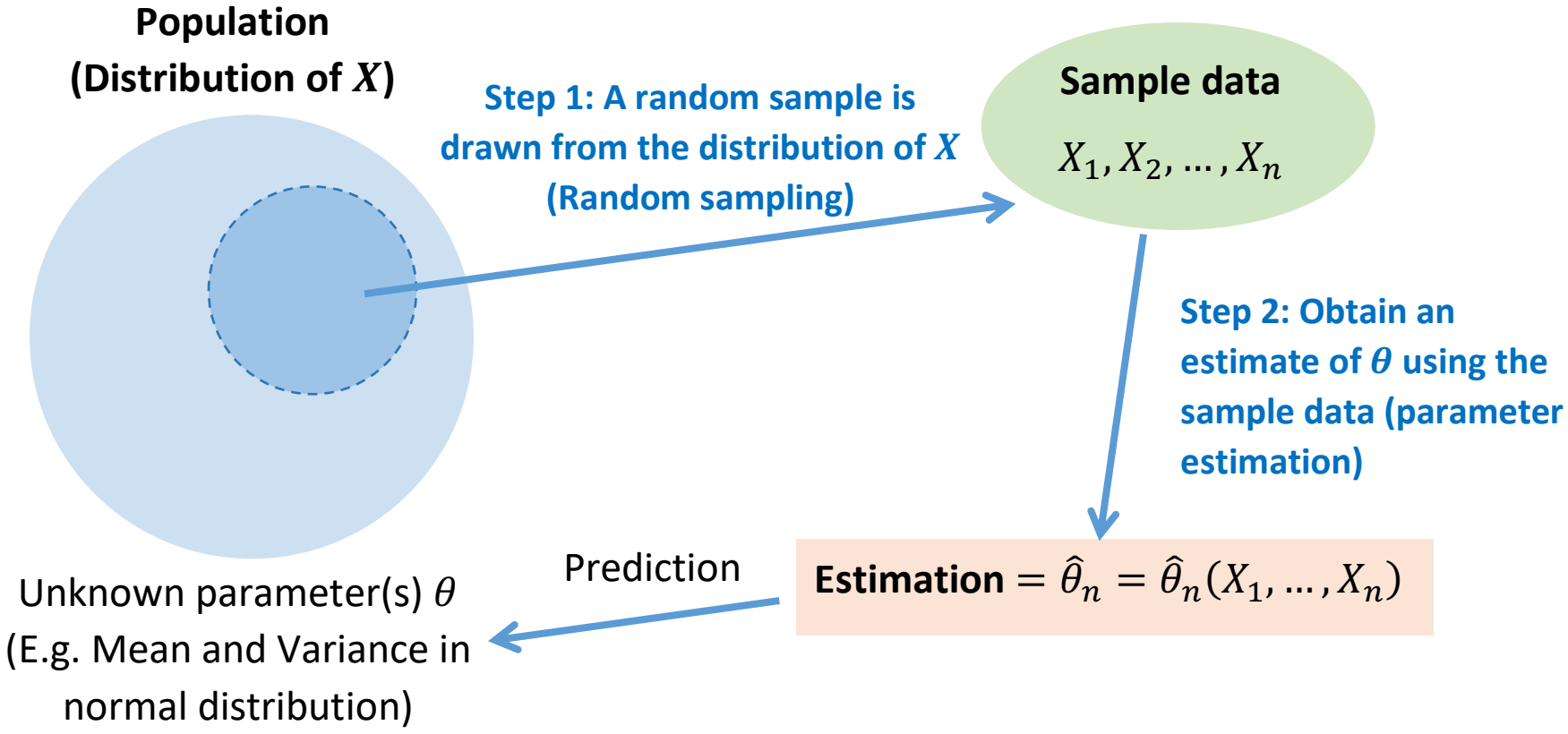
Step 1: Collect the stock data from 2011-2019 and calculate the daily return (or log daily return)

Step 2: Visualize the distribution of stock return using histogram and make a "guess" on the true distribution of stock return.

Step 3: Determine the parameters using the sample data

A quick review on basic theory of parameter estimation

Roughly speaking, parameter estimation aims to predict the unknown parameters in the proposed model. Technically, it aims to fine-tune the model so that the resulting model is the most consistent to the actual data.



The key step of this approach is to obtain a formula for estimating θ, i.e. $\hat{\theta}_n = \hat{\theta}_n(X_1, X_2, \dots, X_n)$. Mathematically, it is called *point estimator* of θ.

Definition (Point Estimator)

We let $\{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a population X and let θ be an unknown parameter associated with the population.

An point estimator $\hat{\theta}_n = \hat{\theta}_n(X_1, X_2, \dots, X_n)$ is a function (i.e. formula) that maps a random sample to a single value.

In general, there are two approaches for obtaining the point estimator:

Method 1: Guess the formula through investigating the nature of the parameters:

Example

If X has normal distribution with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = Var(X)$, then

$$\hat{\mu}_n = \bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n},$$

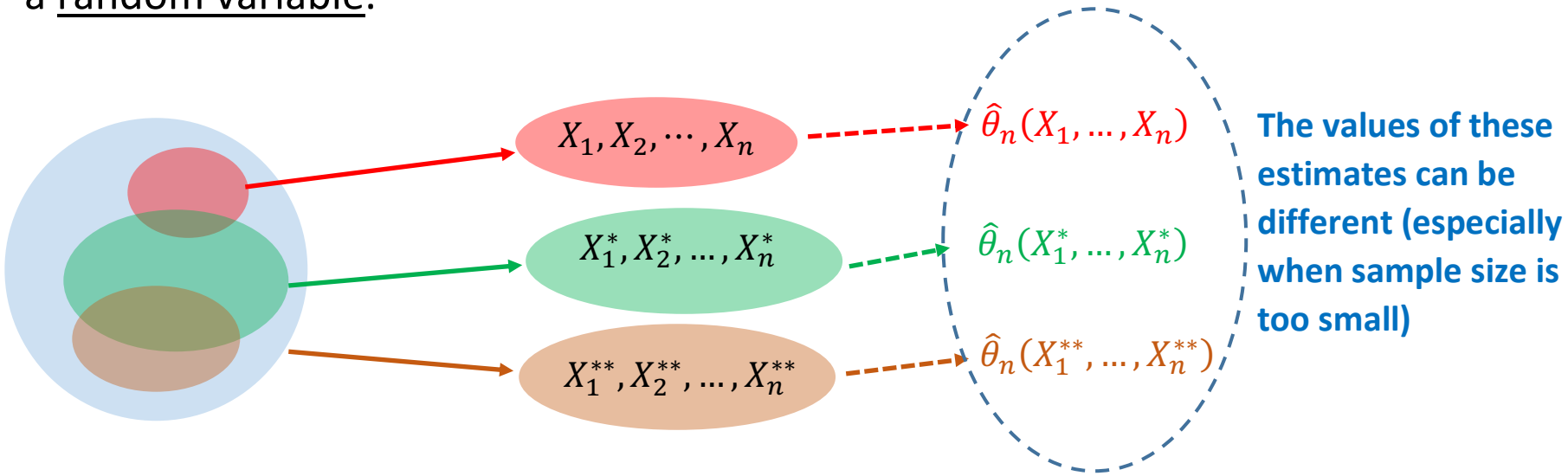
$$\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Method 2: Derive the formula through some standard statistical methods

- *Method of moments* – Derive the formula $\hat{\theta}_n$ through matching the moments $\mathbb{E}[X^k]$.
- *Maximum likelihood method* – Derive the formula by maximizing the *likelihood function*.

Accuracy of the point estimator

Since the value of the estimator is driven by the random sample and the sample is drawn randomly from the population, so there is no doubt that $\hat{\theta}_n$ is a random variable.



Therefore, it is important to ensure that the formula for $\hat{\theta}_n$ can *always* produce an accurate estimate in the sense that $\hat{\theta}_n$ is always close to the true value θ, regardless of the sample data obtained.

Criterions for good point estimators

In classical statistical theory, there are 3 popular criterions that judge if a given point estimator is good or not. Here is a quick summary:

1. **Unbiased estimator**

   We say an estimator $\hat{\theta}_n$ is unbiased if and only if the expected value of $\hat{\theta}_n$ equals to θ. That is,

   $$\mathbb{E}\left[\hat{\theta}_n|\theta\right] = \theta.$$

   Otherwise, we say $\hat{\theta}_n$ is biased and the deviation can be measured by *bias* which is defined as

   $$Bia_\theta\left(\hat{\theta}_n\right) = \mathbb{E}\left[\hat{\theta}_n|\theta\right] - \theta.$$

2. **Consistent estimator**

   We say an estimator $\hat{\theta}_n$ is consistent if and only if the probability that $\hat{\theta}_n$ is close to θ is close to 1 when the sample size n tends to infinity. That is,

   $$\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta\right| > \varepsilon\right) = 0, \quad for\ any\ \varepsilon > 0.$$

3. Lowest mean square error

Mean square error (MSE in short) aims to measure the "exact distance" between $\hat{\theta}_n$ (estimated value) and θ (true value) and is defined as

$$MSE(\hat{\theta}_n) = \mathbb{E}\left[(\hat{\theta}_n - \theta)^2\right]$$

- By some algebra, one can show that

$$MSE(\hat{\theta}_n) = Var(\hat{\theta}_n) + \left(Bia_\theta(\hat{\theta}_n)\right)^2.$$

- Ideally, one should choose an estimator which is (1) *unbiased* and (2) have *lowest mean square error*. If the estimator is unbiased such that $Bia_\theta(\hat{\theta}_n) = 0$, then the MSE can be expressed as

$$MSE(\hat{\theta}_n) = Var(\hat{\theta}_n).$$

  Then the ideal estimator would be the estimator with minimum variance. So we call such estimator as uniformly minimum variance unbiased estimator (UMVUE).

  (*Remark: In practice, it is possible that there is an estimator (not unbiased but have lower mean square error)

Example 1 (Unbiasedness check)

We let X be a random variable which has normal distribution with mean μ and variance σ^2. To estimate μ and σ^2, a random sample of size n is taken from the distribution. Suppose that we estimate μ and σ^2 by

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}, \qquad s^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Determine if each of $\bar{X}$ and s^2 is unbiased estimator.

😊 Solution

Since all X_i are drawn from the population with meanμ and variance σ^2, we have $\mathbb{E}[X_i] = \mu$ and $Var(X_i) = \sigma^2$.

For sample mean $\bar{X}$, we deduce that

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\mu = \frac{n\mu}{n} = \mu.$$

So $\bar{X}$ is unbiased.

For sample variance s^2, we have

$$\mathbb{E}[s^2] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[(X_i - \bar{X})^2]$$

$$= \frac{1}{n}\left(\sum_{i=1}^{n}\mathbb{E}[X_i^2] - 2\sum_{i=1}^{n}\mathbb{E}[X_i\bar{X}] + \sum_{i=1}^{n}\mathbb{E}[\bar{X}^2]\right)$$

$Var[X_i]=E[X_i^2]-(E[X_i])^2$
$\Rightarrow E[X_i^2]=\sigma^2+\mu^2$

$$\overset{\text{☰}}{=} \frac{1}{n}\left[\sum_{i=1}^{n}(\sigma^2+\mu^2) - 2\mathbb{E}\left[\bar{X}\underbrace{\left(\sum_{i=1}^{n}X_i\right)}_{=n\bar{X}}\right] + \sum_{i=1}^{n}\mathbb{E}[\bar{X}^2]\right]$$

$\mathbb{E}[\bar{X}^2]=Var(\bar{X})+(\mathbb{E}[\bar{X}])^2$
$=\frac{\sigma^2}{n}+\mu^2$

$$\overset{\text{☰}}{=} \frac{1}{n}\left[n(\sigma^2+\mu^2) - 2n\left(\frac{\sigma^2}{n}+\mu^2\right) + n\left(\frac{\sigma^2}{n}+\mu^2\right)\right]$$

$$= \frac{1}{n}[(n-1)\sigma^2] = \frac{n-1}{n}\sigma^2 \neq \sigma^2.$$

So s^2 is biased.

Remark of Example 1

We observe from the above example that s^2 is biased for a given n.

- However, one can show that $\lim_{n\to\infty} \mathbb{E}[s^2] = \sigma^2$. It indicates that s^2 is *almost unbiased* when the sample size n is sufficiently large.
  In general, we say an estimator is *asymptotically unbiased* if and only if
  $$\lim_{n\to\infty} \mathbb{E}\left[\hat{\theta}_n|\theta\right] = \theta.$$

- In order to obtain an unbiased estimator for σ^2. One needs to *modify* the formula of the estimator by multiply the original estimator by $\frac{n}{n-1}$.
  $$\hat{s}^2 = \frac{n}{n-1}s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$
  Then one can use the similar calculation and obtain $\mathbb{E}[\hat{s}^2] = \frac{n}{n-1}\mathbb{E}[s^2] = \frac{n}{n-1}\left(\frac{n-1}{n}\sigma^2\right) = \sigma^2$ so that the modified estimator $\hat{s}^2$ is unbiased.

Example 2 (A harder example for unbiasedness check)

A random variable X has uniform distribution on the interval $[0, \theta]$. To estimate θ, a random sample (denoted by $X_1, X_2, \dots, X_n$) is drawn from the distribution.

Suppose that we estimate the parameter θ by $\hat{\theta}_n = \max(X_1, X_2, \dots, X_n)$, determine if $\hat{\theta}_n$ is an unbiased estimator.

(*Note: θ can represent the *maximum possible value* of the random variable X. Thus, one may estimate the value of θ by considering the maximum value of X_i in the random sample)

😊 Solution

Our goal is to compute $\mathbb{E}[\hat{\theta}_n]$. To do so, one needs to obtain the probability density function (p.d.f) of $\hat{\theta}_n$. This can be done as follows:

Note that for any $y \in (0, \theta)$, we have

$$F_{\hat{\theta}_n}(y) = P(\hat{\theta}_n \leq y) = P(\max(X_1, \dots, X_n) \leq y)$$

$$= P(X_1 \leq y, X_2 \leq y, \dots, X_n \leq y) = P(X_1 \leq y)P(X_2 \leq y) \dots P(X_n \leq y)$$

$$\overset{X_1,\ldots,X_n \text{ are iid}}{=} \left(P(X_1 \le y)\right)^n = \left(\int_0^y \frac{1}{\theta}dx\right)^n = \left(\frac{y}{\theta}\right)^n.$$

So the p.d.f. of $\hat{\theta}_n$ can be computed as

$$f_{\hat{\theta}_n}(y) = \frac{d}{dy}F_{\hat{\theta}_n}(y) = \frac{ny^{n-1}}{\theta^n}, \quad for\ y \in (0,\theta).$$

Hence, the expected value of $\hat{\theta}_n$ is given by

$$\mathbb{E}\left[\hat{\theta}_n\right] = \int_0^\theta y f_{\hat{\theta}_n}(y)dy = \int_0^\theta y\left(\frac{ny^{n-1}}{\theta^n}\right)dy = \frac{n}{\theta^n}\left[\frac{y^{n+1}}{n+1}\right]_0^\theta = \frac{n}{n+1}\theta.$$

Since $\mathbb{E}\left[\hat{\theta}_n|\theta\right] \neq \theta$, so $\hat{\theta}_n$ is not unbiased estimator.

Remark of Example 2 (Asymptotically unbiased estimator)

Although $\hat{\theta}_n$ is biased, one can observe that $\lim_{n\to\infty} \mathbb{E}\left[\hat{\theta}_n|\theta\right] = \lim_{n\to\infty} \frac{n}{n+1}\theta = \theta$. So we say $\hat{\theta}_n$ is *asymptotically unbiased* in the sense that it becomes nearly unbiased when the sample size is large.

Example 3 (Example 2 revisited)

Determine if the point estimator $\hat{\theta}_n$ in Example 2 is consistent.

😊 Solution

Recall from Example 2 that the probability density function of $\hat{\theta}_n$ is

$$f_{\hat{\theta}_n}(x) = \frac{nx^{n-1}}{\theta^n}, \quad for\ x \in (0, \theta).$$

Then for any $\varepsilon > 0$ and provided that ε is small such that $\varepsilon < \theta$, we have

$$\lim_{n\to\infty} P\left(\left|\hat{\theta}_n - \theta\right| > \varepsilon\right) \overset{\hat{\theta}_n \le \theta}{=\!=} \lim_{n\to\infty} P\left(\hat{\theta}_n < \theta - \varepsilon\right) = \lim_{n\to\infty} \int_0^{\theta-\varepsilon} \frac{nx^{n-1}}{\theta^n}\, dx$$

$$= \lim_{n\to\infty} \left[\frac{x^n}{\theta^n}\right]_0^{\theta-\varepsilon} = \lim_{n\to\infty} \left(\frac{\theta-\varepsilon}{\theta}\right)^n = 0.$$

So we can conclude that $\hat{\theta}_n$ is consistent by definition.

Remark: When $\varepsilon \ge \theta$, we can deduce directly that $P\left(\hat{\theta}_n < \theta - \varepsilon\right) = 0$ so that

$$\lim_{n\to\infty} P\left(\hat{\theta}_n < \theta - \varepsilon\right) = 0.$$

Example 4 (Consistent estimator)

We let $\{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a distribution X with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = Var(X) < \infty$

(a) Show that the point estimator $\hat{\theta}_1 = \bar{X} = \frac{X_1+X_2+\cdots+X_n}{n}$ is consistent estimator for μ.

(b) Determine if the point estimator $\hat{\theta}_2 = X_n$ (using one sample only) is consistent estimator for μ. Assume that X can take any value over $\mathbb{R}$.

☺Solution of (a)

According to Central limit theorem (as $X_i s$ are i.i.d.), $\bar{X}$ is approximately normal with mean $\mathbb{E}[\bar{X}] = \mu$ and variance $Var(\bar{X}) = \frac{\sigma^2}{n}$ when $n \to \infty$.

Thus for any $\varepsilon > 0$, we have

$$\lim_{n\to\infty} P(|\bar{X} - \mu| > \varepsilon) = \lim_{n\to\infty} [1 - P(|\bar{X} - \mu| \leq \varepsilon)]$$

$$= 1 - \lim_{n\to\infty} P(\mu - \varepsilon \leq \bar{X} \leq \mu + \varepsilon) = 1 - \lim_{n\to\infty} P\left(-\frac{\varepsilon}{\sigma/\sqrt{n}} \leq Z \leq \frac{\varepsilon}{\sigma/\sqrt{n}}\right)$$

$$= 1 - \lim_{n\to\infty} \int_{-\frac{\varepsilon\sqrt{n}}{\sigma}}^{\frac{\varepsilon\sqrt{n}}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = 1 - \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz}_{=1} = 1 - 1 = 0.$$

So $\bar{X}$ is consistent estimator for μ.

☺Solution of (b)

We let $f_X(x)$ be the probability density function of X.

By the similar calculation, we have

$$\lim_{n\to\infty} P(|X_n - \mu| > \varepsilon) = \lim_{n\to\infty} [P(X_n > \mu + \varepsilon) + P(X_n < \mu - \varepsilon)]$$

$$= \lim_{n\to\infty} \left[\int_{\mu+\varepsilon}^{\infty} f_X(x)dx + \int_{-\infty}^{\mu-\varepsilon} f_X(x)dx \right] = \underbrace{\int_{\mu+\varepsilon}^{\infty} f_X(x)dx}_{>0} + \underbrace{\int_{-\infty}^{\mu-\varepsilon} f_X(x)dx}_{>0} \neq 0$$

So $\hat{\theta}_2 = X_n$ is not consistent estimator for μ.

Example 5 (Choosing the best estimator using mean square error)

We consider the setup in Example 2, suppose that we estimate θ by the following two estimators:

$$\hat{\theta}_A = 2\bar{X} \quad and \quad \hat{\theta}_B = \frac{n+1}{n}\max(X_1, X_2, \dots, X_n),$$

Compare the qualities of two estimators by computing their MSEs.

☺Solution

First, one can verify that two estimators are unbiased (so $Bia_\theta(\hat{\theta}) = 0$), i.e.,

$$\mathbb{E}[\hat{\theta}_A|\theta] = 2\mathbb{E}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right] = \frac{2}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] \overset{\mathbb{E}[X_i]=\int_0^\theta \frac{x}{\theta}dx=\frac{x^2}{2\theta}|_0^\theta=\frac{\theta}{2}}{=} \frac{2}{n}\sum_{i=1}^{n}\left(\frac{\theta}{2}\right)$$

$$= \frac{2}{n}\left(\frac{n\theta}{2}\right) = \theta.$$

$$\mathbb{E}[\hat{\theta}_B|\theta] = \frac{n+1}{n}\mathbb{E}[\max(X_1, X_2, \dots, X_n)] = \frac{n+1}{n}\left(\frac{n}{n+1}\theta\right) = \theta.$$

It remains to calculate the variance $Var\left(\hat{\theta}\middle|\theta\right)$.

Since X_i follows uniform distribution, we have

$$\mathbb{E}[X_i] = \frac{\theta}{2}, \quad \mathbb{E}\left[X_i^2\right] = \int_0^\theta x^2 \left(\frac{1}{\theta}\right) dx = \left[\frac{x^3}{3\theta}\right]_0^\theta = \frac{\theta^2}{3}$$

$$Var(X_i) = \mathbb{E}\left[X_i^2\right] - (\mathbb{E}[X_i])^2 = \frac{\theta^2}{3} - \left(\frac{\theta}{2}\right)^2 = \frac{\theta^2}{12}.$$

The variance of $\hat{\theta}_A$ is given by

$$Var\left(\hat{\theta}_A\right) = Var\left(2\frac{X_1 + X_2 + \cdots + X_n}{n}\right) \overset{Var(cX)=c^2Var(X)}{=} \frac{4}{n^2}\sum_{i=1}^{n} Var(X_i)$$

$$= \frac{4}{n^2}\left(\frac{n\theta^2}{12}\right) = \frac{\theta^2}{3n}.$$

So we have $MSE_{\hat{\theta}_A}(\theta) = Var\left(\hat{\theta}_A\right) + \left(Bia_\theta\left(\hat{\theta}_A\right)\right) = \frac{\theta^2}{3n}$.

On the other hand, the second moment of $\hat{\theta}_B$ is given by

$$\mathbb{E}\left[\hat{\theta}_B^2|\theta\right] = \int_0^\theta \left(\frac{n+1}{n}y\right)^2 \frac{ny^{n-1}}{\theta^n} dy = \frac{(n+1)^2}{n\theta^n}\int_0^\theta y^{n+1} dy$$
$$= \frac{(n+1)^2}{n\theta^n}\left[\frac{y^{n+2}}{n+2}\right]_0^\theta = \frac{(n+1)^2}{n(n+2)}\theta^2.$$

Hence, the variance of $\hat{\theta}_B$ is found to be

$$Var\left(\hat{\theta}_B\middle|\theta\right) = \mathbb{E}\left[\hat{\theta}_B^2|\theta\right] - \left(\mathbb{E}\left[\hat{\theta}_B|\theta\right]\right)^2 = \frac{(n+1)^2}{n(n+2)}\theta^2 - \theta^2 = \frac{\theta^2}{n(n+2)}.$$

So $MSE_{\hat{\theta}_B}(\theta) = \frac{\theta^2}{n(n+2)}$.

Since $n \geq 1$, we must have

$$MSE_{\hat{\theta}_B}(\theta) = \frac{\theta^2}{n(n+2)} \leq \frac{\theta^2}{n(1+2)} = \frac{\theta^2}{3n} = MSE_{\hat{\theta}_A}(\theta).$$

So $\hat{\theta}_B$ is better than $\hat{\theta}_A$ since it has smaller MSE.

Interval estimation and confidence interval

The point estimator $\hat{\theta}_n$ can provide a reasonable estimate on the value of the parameter value θ (provided that the sample size is large). However, the point estimate may still deviate from the true value θ so that $\hat{\theta}_n$ may not reflect too much information about θ.

Instead, one may consider finding the *range* of θ which may possibly reveal more information about the true value of θ.

- Given sample data $\{X_1, X_2, \dots, X_n\}$, we determine $\hat{\theta}_L = \hat{\theta}_L(X_1, \dots, X_n)$ and $\hat{\theta}_U = \hat{\theta}_U(X_1, X_2, \dots, X_n)$ such that $\hat{\theta}_L \leq \theta \leq \hat{\theta}_U$.

Note that the values of $\hat{\theta}_L$ and $\hat{\theta}_U$ are determined through the random sample and there is no 100% guarantee that such prediction is correct. So we would like to make a prediction to be true for a high level of probability. That is,

$$P\left(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U\right) = p,$$

where p is a probability which takes a fairly large value (such as 90%, 95%, 98% even 99%). The interval $[\theta_L, \theta_U]$ is known as $(100p)$% confidence interval.

How to obtain the confidence interval of an estimator? A general methodology

To understand the procedure, we consider a classical example which we determine the 95% confidence interval of population mean μ from a population with known variance σ^2.

We let $\{X_1, X_2, \dots, X_n\}$ be the random sample obtained and let $\bar{X} = \frac{X_1+X_2+\cdots+X_n}{n}$ be the sample mean, which is a point estimator for μ.

To obtain the confidence interval for μ, we note that

- Provided that the sample size is fairly large, one can deduce from Central limit theorem that $\bar{X}$ follows normal distribution with mean μ and variance $\frac{\sigma^2}{n}$ (i.e. $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$). Equivalently, we have

$$Z = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim N(0,1).$$

- Given the distribution of Z, we obtain the thresholds z_L and z_U such that

$$P\left(z_L \leq Z = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \leq z_U\right) = 95\% \dots (*)$$

  It is clear that there are infinitely many possible choices for (z_L, z_U) that satisfy the equation. As a common practice, we choose z_L and z_U such that (why?)

$$P(Z \leq z_L) = P(Z \geq z_U) = \frac{0.05}{2} \Rightarrow z_L = -1.96 \ \ and \ \ z_U = 1.96.$$

- Substitute the result in the equation $(*)$, we get

$$P\left(-1.96 \le Z = \frac{\bar{X}-\mu}{\sqrt{\sigma^2/n}} \le 1.96\right) = 95\% \Rightarrow P\left(\underbrace{\bar{X} - 1.96\sqrt{\frac{\sigma^2}{n}}}_{\hat{\theta}_L} \le \mu \le \underbrace{\bar{X} + 1.96\sqrt{\frac{\sigma^2}{n}}}_{\hat{\theta}_U}\right) = 95\%.$$

  Then $\left[\hat{\theta}_L, \hat{\theta}_U\right]$ is the desired confidence interval for μ in the sense that the prediction is correct with 95% probability.

(*Remark: In practice, the population variances is also unknown and one needs to modify the approach to obtain the correct confidence interval. See Example ??)

The procedure can be extended to find the confidence interval for general parameter θ.

- Step 1: Obtain a point estimator for θ (denoted by $\hat{\theta}_n$).
- Step 2: Determine the probability distribution for $\hat{\theta}_n$ (either by ad-hoc method or approximation theory for large sample)
- Step 3: Find the constants L and U such that $P\left(L \le f\left(\hat{\theta}_n, \theta\right) \le U\right) = p$.
- Step 4: Solve the inequality $L \le f\left(\hat{\theta}_n, \theta\right) \le U$ for θ to obtain the required confidence interval. That is, $P\left(\hat{\theta}_L \le \theta \le \hat{\theta}_U\right) = p$.

Example 6

In this example, we would like to examine the probability that daily price change of a given stock (Tencent, 0700.HK) exceeds 3%. To do so, we analyse the daily return of the stock from 1st Jan, 2011 to 31st Dec, 2019 (sample size = 2213). We let p be the probability that daily price change in a given day exceeds 3%. Then p is estimated to be

$$\hat{p} = \frac{number\ of\ data\ with\ return < -3\%\ or\ return > 3\%}{number\ of\ data} = \frac{253}{2213} \approx 0.1156801.$$

Determine the 95% *symmetric* confident interval for p.

*Note: In R-Studio, one can obtain the estimates using the following commands:

```
X0700_HK<-na.omit(X0700_HK) #Remove the N.A. entries from the dataset
x<-X0700_HK$Return #Take out the "Return" data and store in "x"

N<-sum(ifelse(abs(x)>0.03,1,0)) #Count the number of data with return<-3% or >3% and s
tore in "N"
n<-length(X0700_HK$Return) #Calculate sample size n

estimate=N/n #Calcualte the estimate of p

>print(N)
[1] 256
> print(n)
[1] 2213
> print(estimate)
[1] 0.1156801
```

☺Solution

We first determine the distribution of the estimator $\hat{p}$. Note that $\hat{p}$ can be expressed as

$$\hat{p} = \frac{X_1 + X_2 + \cdots + X_{2213}}{2213},$$

where $X_i = \begin{cases} 1 & if \ \ price\ change\ in\ i^{th} day > 3\% \\ 0 & if \ \ price\ change\ in\ i^{th} day < 3\% \end{cases}$ and $P(X_i = 1) = p$.

By central limit theorem, $\hat{p}$ have normal distribution with mean $\mathbb{E}[\hat{p}] = \frac{2213\mathbb{E}[X_i]}{2213} = \frac{2213p}{2213} = p$

and $Var(\hat{p}) = \frac{2213Var(X_i)}{(2213)^2} \overset{Var(X_i)=\mathbb{E}[X_i^2]-(\mathbb{E}[X_i])^2}{=\!=\!=} \frac{p(1-p)}{2213}$.

Next, we find the numbers z_L and z_U such that

$$P\left(z_L \leq \frac{\hat{p} - \mathbb{E}[\hat{p}]}{\sqrt{Var(\hat{p})}} \leq z_U\right) = 0.95.$$

In usual practice, we consider symmetric confidence interval and choose z_L and z_U such that $P(Z \leq z_L) = P(Z \geq z_U) = 0.025$.

This yields $z_L = -1.96$ and $z_U = 1.96$.

So the above inequalities becomes

$$P\left(-1.96 \leq \frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{2213}}} \leq 1.96\right) = 0.95$$

$$\Rightarrow P\left((\hat{p}-p)^2 \leq (1.96)^2 \frac{p(1-p)}{2213}\right) = 0.95$$

$$\Rightarrow P\left(\overbrace{1.001736}^{a} p^2 - \left(\overbrace{2\hat{p} + 0.001735924}^{b}\right)p + \overbrace{\hat{p}^2}^{c} \leq 0\right) = 0.95.$$

$$\Rightarrow P\left(\frac{b-\sqrt{b^2-4ac}}{2a} \leq p \leq \frac{b+\sqrt{b^2-4ac}}{2a}\right)$$

(*Note: Last equality follows from quadratic formula)

Using $\hat{p} = 0.1156801$, we deduce that the 95% confidence interval of p is $[0.101776,\ 0.128313]$.

Example 7 (Finding confidence interval using R)

Suppose that we are interested in finding 98% confidence intervals for average daily return (denoted by μ) and variance of daily return (denoted by σ^2) of HSBC stock (0005.HK). We assume that the daily return is normally distributed.

Analysis method

To do so, we collect the prices of the stock over the period from 1st Jan, 2011 to 31st Jul, 2019. Next, we compute the daily return of the stock using *adjusted closing price* (i.e. the stock price after the deduction of dividends). Here, the sample size is $n = 2213$.

The sample mean and sample variance are found to be

$$\bar{r} = 8.186392 \times 10^{-5}, \qquad s = 0.01214441.$$

☺Solution

Given that the log-return is normally distributed by the assumption, we have

- ✓ $\bar{r}$ is normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.
- ✓ $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$ has chi-squared distributed with degree of freedom $n - 1$
- ✓ $\bar{r}$ and s^2 are independent.

Then one can show that the statistics

$$T = \frac{\bar{r} - \mu}{s/\sqrt{n}} = \frac{\left(\frac{\bar{r} - \mu}{\sigma/\sqrt{n}}\right)}{\sqrt{s^2/\sigma^2}} = \frac{Z}{\sqrt{\chi^2/(n-1)}}$$

follows student-t distribution with degree of freedom $n - 1 = 2212$ (see Remark).

Remark (Definition of chi-square distribution and student-t distribution

- We let $Z_1, Z_2, \dots, Z_n$ be n independent standard normal random variable. We say the random variable defined by

  $$\chi_n^2 = Z_1^2 + Z_2^2 + \cdots + Z_n^2$$

  has chi-squared distribution with degree of freedom n.
  (The distribution can be extended to the case when n is positive real number)
- We let Z be a standard normal random variable and χ_n^2 be a chi-squared random variable with degree of freedom n. Z and χ_n^2 are independent, the random variable defined by

  $$T_n = \frac{Z}{\sqrt{\chi_n^2/n}}$$

  has student-t distribution with degree of freedom n.

Next, we proceed to find the thresholds t_L and t_U such that

$$P(t_L \leq T \leq t_U) = 0.98.$$

By choosing $P(T \leq t_L) = P(T \geq t_U) = 0.01$, we have $t_U = 2.328035$ and $t_L = -2.328035$. Thus, we deduce from equation $(*)$ that

$$P\left(-2.328035 \leq \frac{\bar{r} - \mu}{s/\sqrt{n}} \leq 2.328035\right) = 0.98$$

$$\Rightarrow P\left(\bar{r} - 2.328035\frac{s}{\sqrt{n}} \leq \mu \leq \bar{r} + 2.328035\frac{s}{\sqrt{n}}\right) = 0.98.$$

Using the estimates obtained from the data, the 98% confidence interval for average return μ is $[-0.005191371, 0.000682865]$.

Remark of Example 7

- In R, one can determine the *quantile* of chi-squared distribution and T-distribution by the following commands:
  - For student-t distribution, "qt(p, n)" generates T_0 which satisfies $P(T \leq T_0) = p$,
  - For chi-squared distribution, "qchisq(p, n)" generates χ_0^2 which satisfies $P(\chi^2 \leq \chi_0^2) = p$.

Parameter estimation methods in parametric models

Suppose that a random variable X has a distribution with parameter θ and we would like to estimate θ using the sample data.

Besides simple guess as demonstrated in , there are some systematic methods for finding the point estimator $\hat{\theta}_n$ of θ, namely method of moments and maximum likelihood method.

Method 1: Method of moments

The principle of this method can be understood as follows:

Suppose that a random variable X has a distribution with parameters θ, then the corresponding moments (i.e. $\mathbb{E}[X^k]$) should be functions of θ.

- If X has normal distribution with mean μ and variance σ^2, then we have $\mathbb{E}[X] = \mu$ and $\mathbb{E}[X^2] = \mu^2 + \sigma^2$.
- If Y has exponential distribution with mean $\theta > 0$, then $\mathbb{E}[X] = \theta$ and $\mathbb{E}[X^k] = \theta^k k!$ for any $k = 1,2,3, \ldots$

According to strong law of large number, one can estimate the kth moments using sample data by

$$\mathbb{E}[X^k] \approx \frac{X_1^k + X_2^k + \cdots + X_n^k}{n}.$$

Then, one can deduce governing equation(s) for the unknown parameter(s) and obtain the estimate(s) by solving the equations.

Example 8

We consider the logreturn of Tencent stock (denoted by X) and suppose that we would like to model the stock logreturn by normal distribution with mean μ and variance σ^2. Determine μ and σ^2 using method of moments.

☺Solution

Using the sample data, we get

$$\frac{X_1 + X_2 + \cdots + X_n}{n} \approx 0.001089745, \quad \frac{X_1^2 + X_2^2 + \cdots + X_n^2}{n} \approx 0.0003827469.$$

Using the fact that $\mathbb{E}[X] = \mu$ and $\mathbb{E}[X^2] = \mu^2 + \sigma^2$, we deduce that

$$\begin{cases} \mu = \mathbb{E}[X] \approx \dfrac{X_1 + X_2 + \cdots + X_n}{n} \approx 0.001089745 \\ \mu^2 + \sigma^2 = \mathbb{E}[X^2] = \dfrac{X_1^2 + X_2^2 + \cdots + X_n^2}{n} \approx 0.0003827469 \end{cases} \Rightarrow \begin{cases} \mu = 0.001089745 \\ \sigma = 0.01953354 \end{cases}$$

(*Here, we keep the positive root for σ since $\sigma > 0$ by default)

Example 9 (An experiment on method of moments)

A random variable X has exponential distribution with mean θ (i.e. the probability density function is $f_X(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$, $x > 0$ and $\mathbb{E}[X^k] = \theta^k k!$).

To estimate θ, a random sample of $n = 20000$ is drawn from the distribution of X. Estimate the value of θ by considering the 1st moment, 2nd moment and 3rd moment respectively.

☺Solution

Given a fixed positive integer k, θ can be estimated as

$$\theta^k k! = \mathbb{E}[X^k] \approx \frac{\sum_{i=1}^n X_i^k}{n} \Rightarrow \theta \approx \left(\frac{\sum_{i=1}^n X_i^k}{n(k!)}\right)^{\frac{1}{k}}.$$

One can perform the calculation through the following r-code: (In this experiment, we take the true value to be $\theta = 2$)

```
N=20000 #Sample size
theta=2 #True value of theta
rate=1/theta

x<-rexp(N,rate) #Generate random sample from exponential distribution

#Define a function for getting MME using sample and given information
MME<-function(k,x){
  n=length(x)
  estimate=(sum(x^k)/(n*factorial(k)))^(1/k)
  return(estimate)
}

MME1=MME(1,x) #Calculate the estimate for k=1
MME2=MME(2,x) #Calculate the estimate for k=2
MME3=MME(3,x) #Calculate the estimate for k=3
```

We summarize the results in the following table:

	Simulation 1	Simulation 2	Simulation 3
$k = 1$	$\hat{\theta}_n = 2.029646$	$\hat{\theta}_n = 2.002628$	$\hat{\theta}_n = 1.980935$
$k = 2$	$\hat{\theta}_n = 2.030453$	$\hat{\theta}_n = 2.00124$	$\hat{\theta}_n = 1.981887$
$k = 3$	$\hat{\theta}_n = 2.028842$	$\hat{\theta}_n = 1.998358$	$\hat{\theta}_n = 1.976134$

(*The estimates varies in different trials because different samples may be drawn in different runs)

- ✓ We observe from the above table that different choices of k appears to yield the estimates with similar accuracy. This shows that the choice of k is not a crucial issue when using method of moments.
- ✓ Therefore, it is more convenient to choose small value of k when we use this method.
  - It is easier to obtan $\mathbb{E}[X^k]$ for small value of k.
  - For distribution with heavy tail, $\mathbb{E}[X^k]$ may not exists as number for large value of k.

Limitation of method of moments

Although method of moments allow us to obtain a point estimator for θ easily, it is possible that it can yield an estimate which is inconsistent to the sample data. To see this, we consider the following example:

A random variance X has uniform distribution over $[0, \theta]$, where θ is an unknown parameter. Suppose that the random sample is given as follows:

X_i	1	2	3	100
Frequency	3506	3391	3500	3

Using method of moment, θ can be estimated as

$$\frac{\theta}{2} = \mathbb{E}[X] \approx \frac{1}{10400} \sum_{i=1}^{10400} X_i \Rightarrow \theta = 4.0554.$$

We see that such estimate is not consistent with actual data in the sense that there are 3 data greater than θ. It is because the method of moments consider the mean of the sample data which the value is controlled by majority of data and may ignore the presence of some extreme data since those data do not contributed much to the mean.

Method 2 – Maximum Likelihood Method

Maximum likelihood method is an estimation method which uses *all information* available in the sample data. More precisely, this method tries to estimate the value of the parameters by considering the *probability* that the sample data is observed given a set of parameter values.

Likelihood function

We let X be a random variable which has probability distribution with unknown parameters $\theta_1, \theta_2, \dots, \theta_k$. A random sample of size n (denoted by $X_1, X_2, \dots, X_n$) is drawn from the distribution.

To estimate the parameters $\theta_1, \theta_2, \dots, \theta_k$, we consider the probability that the random sample $\{X_1, X_2, \dots, X_n\}$ is obtained. That is,

$$L(\theta_1, \theta_2, .., \theta_k) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n | \theta_1, \dots, \theta_k)$$
$$= \prod_{i=1}^{n} P(X_i = x_i | \theta_1, \theta_2, \dots, \theta_k).$$

The function $L(\theta_1, \theta_2, \dots, \theta_k)$ is called *likelihood function*.

What values of $\theta_i's$ should we choose?

Given the fact that $(X_1, X_2, \dots, X_n) = (x_1, x_2, \dots, x_n)$ is observed in reality, one should expect that such observation should occur with very high probability. So we choose the parameter $\hat{\theta}_1, \hat{\theta}_2, \dots . \hat{\theta}_k$ such that the above likelihood function is maximized, i.e.

$$\left(\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k\right) = \underset{\theta_1, \theta_2, \dots, \theta_k}{\operatorname{argmax}} L(\theta_1, \theta_2, \dots, \theta_k)$$

The vector $\left(\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k\right)$ is called *maximum likelihood estimate* (MLE) of $(\theta_1, \theta_2, \dots, \theta_k)$.

Remark about maximum likelihood function

- We let $f_X(x)$ be probability density/mass function of X, then the likelihood function can be expressed as

$$L(\theta_1, \theta_2, .., \theta_k) = \prod_{i=1}^{n} f_X(x|\theta_1, \theta_2, \dots, \theta_k)$$

Example 10

(a) A random variable N has Poisson distribution with the probability mass function

$$P(N = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0,1,2, \ldots.$$

where $\lambda > 0$ is a parameter.

Suppose that a random sample $\{N_1, N_2, \ldots, N_n\}$ is drawn from the distribution of N, determine the MLE of λ.

(b) A random variable X has exponential distribution with the probability density function

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x > 0.$$

We let $\{X_1, X_2, \ldots, X_n\}$ be the random sample drawn from the distribution of X, determine the MLE of λ.

☺Solution of (a)

First, the likelihood function is given by

$$L(\lambda) = \prod_{i=1}^{n} \underbrace{\frac{\lambda^{N_i} e^{-\lambda}}{(N_i)!}}_{P(N=N_i)} = \frac{\lambda^{\sum_{i=1}^n N_i} e^{-n\lambda}}{\prod_{i=1}^n (N_i!)}.$$

To find the MLE, we consider the first order condition:

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow \frac{(\sum_{i=1}^n N_i)\lambda^{\sum_{i=1}^n N_i - 1} e^{-n\lambda} - n\lambda^{\sum_{i=1}^n N_i} e^{-n\lambda}}{\prod_{i=1}^n (N_i!)} = 0 \overset{\lambda>0}{\Rightarrow} \hat{\lambda}_n = \frac{\sum_{i=1}^n N_i}{n}.$$

One can verify that $\frac{\partial^2 L}{\partial \lambda^2}|_{\lambda=\hat{\lambda}_n} < 0$. So the MLE of λ is $\hat{\lambda}_n = \frac{\sum_{i=1}^n N_i}{n}$.

☺Solution of (b)

The corresponding likelihood function is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda X_i} = \lambda^n e^{-\lambda(\sum_{i=1}^n X_i)}$$

Next, we consider the first order condition:

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow n\lambda^{n-1} e^{-\lambda(\sum_{i=1}^n X_i)} - \left(\sum_{i=1}^{n} X_i\right) \lambda^n e^{-\lambda(\sum_{i=1}^n X_i)} = 0 \Rightarrow \hat{\lambda}_n = \frac{n}{\sum_{i=1}^n X_i}.$$

Hence, the MLE of λ is $\hat{\lambda}_n = \frac{n}{\sum_{i=1}^n X_i}$.

Remarks (The use of loglikelihood function)

To simplify the calculation, one would consider the loglikelihood function defined by

$$l(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f_X(X_i|\theta).$$

Since $\ln x$ is increasing function in x, so maximizing $l(\theta)$ is equivalent to maximizing $L(\theta)$. As an example, we revisited the calculation in (b):

The loglikelihood function is found to be

$$l(\lambda) = \ln L(\lambda) = \cdots = n \ln \lambda - \lambda \sum_{i=1}^{n} X_i.$$

By considering the first order condition, we deduce that

$$\frac{dl}{d\lambda} = 0 \Rightarrow \frac{n}{\lambda} - \sum_{i=1}^{n} X_i = 0 \Rightarrow \hat{\lambda}_n = \frac{n}{\sum_{i=1}^n X_i}.$$

We obtained the same MLE. However, the calculation is much easier.

Example 11

A random variable X has uniform distribution over $[0, \theta]$. To estimate θ, a random sample of size n $(X_1, X_2, \dots, X_n)$ is drawn from the distribution. Find the maximum likelihood estimate of θ.

😊 Solution

Firstly, the likelihood function (for any $\theta > 0$) is given by

$$L(\theta|X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} f_X(X_i|\theta) = \begin{cases} \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n} & if \ \ \theta \geq \max(X_1, X_2, \dots, X_n) \\ 0 & if \ \ \theta < \max(X_1, X_2, \dots, X_n) \end{cases}.$$

(*Note: Recall that $f_X(X_i|\theta) = \frac{1}{\theta} \ \ if \ \ X_i \in [0, \theta]$.)

For $\theta \geq \max(X_1, X_2, \dots, X_n)$, we observe that $L(\theta)$ is decreasing at θ. Therefore, we can deduce that $L(\theta)$ is maximized at $\theta = \max(X_1, X_2, \dots, X_n)$.

So the MLE of θ is $\hat{\theta}_n = \max(X_1, \dots, X_n)$.

Finding MLEs using RStudio

In RStudio, there are several ways to determine MLE using the sample data, depending on which model distribution is adopted.

Method 1: Use "fitdistr"

For some common distributions (e.g. normal, exponential, Poisson etc.), one can obtain the MLE using the command "fitdistr" (available in "MASS" library)

```
> library(MASS)
> fitdistr(x,"normal")
      mean              sd
  2.02373311    3.00938597
 (0.04255914) (0.03009386)
```

Note:

This approach works well with distributions which has explicit formula for MLE(s).

Method 2: Ad-hoc Searching

Given the likelihood function $L(\theta)$ (or loglikelihood function $l(\theta)$), we compute the numerical values of $L(\theta)$ for a set of sample values of θ over the domain of θ and pick the θ (i.e. MLE) that yields the maximum value of $L(\theta)$.

Note:

This method is applicable to any types of distributions. However, the computational time is much longer.

Method 3

For some special distributions such as multivariate normal distribution or multivariate-t distribution, one can obtain the MLEs through the specific command:

- ✓ "mvnorm.mle(x)" can be used to the MLEs of multivariate normal distribution.
- ✓ "mvt.mle(x,v)" can be used to find the MLEs of multivariate-t distribution, given a value of $\nu > 0$

(*Package "Rfast" is required)

Example 12 (Quick example using "fitdistr")

(a) A random sample of size $n = 10000$ is drawn from the exponential distribution ($f_X(x) = \lambda e^{-\lambda x}, x > 0$) with $\lambda = 3$.
Suppose that we would like to fit the data by exponential distribution, one can determine the MLE as follows:

```
> library(MASS) #Package 'MASS' is required for using fitdistr
> n=10000
> xsample<-rexp(n,3)
>
> fitdistr(xsample,'exponential')
      rate
 3.01353653
(0.03013537)
```

MLE of λ (Here, rate = λ)

Standard Error of the MLE

(b) Suppose that we would like to fit the log-return of the stock 0700.HK by normal distribution with mean μ and variance σ^2. Similar to (a), one can obtain the MLEs of μ and σ as follows:

```
> library(MASS)
> X0700_HK<-na.omit(X0700_HK)
>
> fitdistr(X0700_HK$Logreturn,'normal') #Find the MLEs of parameters (no
rmal distribution)
       mean             sd
  0.0010897454   0.0195335436
 (0.0004152315) (0.0002936130)
```

Example 13 (Finding MLE through ad-hoc searching)

We let X be a random variable which has student-t distribution with degree of freedom $\nu > 0$. That is, the probability density function of X is

$$f_X(x|\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}},$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1}e^{-t}dt$ denotes Gamma function.

A random sample of size $n = 5000$ is drawn (take the true value to be $\nu = 3$ in this example), determine the MLE of ν using this sample data.

☺Solution

There is no explicit formula for MLE of ν (possibly because the unknown parameter is contained in the Gamma function).

Instead, we shall determine the MLE using ad-hoc searching directly. Here is the sample R-code for the execution:

```
x<-rt(5000,3) #Generate syntheic sample from student-t distribution

#Define a function for calculating the loglikelihood function of student-
t distribution (given v)
logt<-function(x,v){
  logfun=log(gamma((v+1)/2))-(1/2)*log(pi*v)-log(gamma(v/2))-((v+1)/2)*lo
g(1+(x^2)/v)
  loglikelihoodt=sum(logfun)
  return(loglikelihoodt)
}

v=seq(0.001,10,0.001) #Generate a list of sample value of v (from 0 to 1
0, excluding 0)

m=length(v)
loglik=rep(0,m) #Define an array to store the sample values of loglikelih
ood function

for (i in 1:m){
  loglik[i]=logt(x,v[i]) #For each v, calculate the sample value of logli
kelihood function
}

imax=which.max(loglik) #Find the entry with maximum loglikelihood value
MLEv=v[imax] #Quote the corresponding value of v as MLE

> MLEv
[1] 3.137
```

Example 14

We let X be a random variable which has gamma distribution with the following probability density function:

$$f_X(x) = \frac{(\lambda x)^{\alpha} e^{-\lambda x}}{x\Gamma(\alpha)}, \qquad x > 0$$

where $\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt$, $\alpha > 0$ and $\lambda > 0$ are unknown parameters.

Suppose that a random sample of size $n = 20000$ is drawn the distribution, determine the MLEs of α and λ using ad-hoc approach.

☺Solution

Although we can get the MLEs using direct ad-hoc searching (i.e. calculating the values of loglikelihood function for different pairs of (α, θ)), the calculation could be quite time-consuming due to curse of dimensionality.

To enhance the efficiency of the algorithm, one can adopt the following approach:

Step 1: Given the value of α, we derive the formula of MLE of λ (denoted by λ_α) which the explicit formula is available:

✓ The loglikelihood function is given by

$$l(\alpha,\lambda)=\sum_{i=1}^{n}\ln f_X(X_i|\alpha,\lambda)=\sum_{i=1}^{n}(\alpha\ln\lambda+(\alpha-1)\ln X_i-\ln\Gamma(\alpha)-\lambda X_i)$$

$$=n\alpha\ln\lambda-(\alpha-1)\sum_{i=1}^{n}\ln X_i-n\ln\Gamma(\alpha)-\lambda\sum_{i=1}^{n}X_i.$$

By considering the first order condition on λ (recall that α is fixed), we get

$$\frac{\partial l}{\partial\lambda}=0\Rightarrow\frac{n\alpha}{\lambda}-\sum_{i=1}^{n}X_i=0\Rightarrow\lambda_\alpha=\frac{n\alpha}{\sum_{i=1}^{n}X_i}.$$

Step 2: Then we can find the MLEs as follows:

(a) For each $\alpha>0$, we compute λ_α and the loglikelihood value $l(\alpha,\lambda_\alpha)$.

(b) Then we choose α such that $l(\alpha,\lambda_\alpha)$ is maximized. We let $\hat{\alpha}$ be the maximizer, then the pair $(\hat{\alpha},\lambda_{\hat{\alpha}})$ will be the MLEs of α,λ.

Here is the sample R-code for this algorithm. (*Note: We take $\alpha = 3$ and $\lambda = 1$ be the true value of the parameters)

```
lambdatrue=1 #Assume the true value of lambda is 1
alphatrue=3 #Assume the true value of alpha is 3
n=20000
x<-rgamma(n,shape=alphatrue,rate=lambdatrue) #Generate random sample

#Define a function for calculating the loglikihood function of gamma distribution
loglikgamma<-function(x,alpha,lambda){
   logfun=alpha*log(lambda)+(alpha-1)*log(x)-lambda*x-log(gamma(alpha))
   return(sum(logfun))
}

alpha=seq(0.001,10,0.001) #Set the range of alpha (must contain the true value)
m=length(alpha)
lambda=rep(0,m)
loglik=rep(0,m) #Array to store the values of loglikelihood for different alpha

#Calculate the loglikelihood value for different values of alpha
for(i in 1:m){
   lambda[i]=n*alpha[i]/sum(x) #Calculate the MLE of lambda given alpha
   loglik[i]=loglikgamma(x,alpha[i],lambda[i])
}

#Obtain the MLE of alpha
t=which.max(loglik) #Finding the index
MLEalpha=alpha[t] #Finding the corresponding value of alpha (MLE)
MLElambda=lambda[t] #Finding the corresponding value of lambda (MLE)

print(MLEalpha)
[1] 2.947
print(MLElambda)
[1] 0.9871804
```

Further remarks about the MLE

So far, there are 2 methods for finding MLEs and it appears that maximum likelihood method is better since it makes use of all information available in the data. Next, we would like to know if this method is a good in the sense that

1. Is MLE a good estimator?
2. Is there any method that can perform better than maximum likelihood method?

We let $\hat{\theta}_n$ be the maximum likelihood estimate of θ satisfying $l'(\hat{\theta}_n) = 0$. Provided that certain *regularity conditions* (which is technical) hold, one can establish the following properties for $\hat{\theta}_n$:

1. $\hat{\theta}_n$ is *asymptotically unbiased* (i.e. $\lim_{n\to\infty} \mathbb{E}[\hat{\theta}_n] = \theta$)
   (*However, $\hat{\theta}_n$ may not be unbiased.)
2. $\hat{\theta}_n$ is *consistent* (i.e. $\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0$)

3. When $n \to \infty$, $\hat{\theta}_n$ has approximately normal distribution with mean θ and variance $\frac{1}{I(\theta)}$, where $I(\theta)$ is called Fisher's information and is defined as

$$I(\theta) = -n\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ln f_X(X|\theta)\right].$$

4. When $n \to \infty$, the MLE has the <u>smallest mean square error</u> among all unbiased estimators. That is, for any other *unbiased* estimator $\hat{\theta}$, we have

$$\underbrace{Var(\hat{\theta})}_{MSE\ of\ \hat{\theta}} \geq \underbrace{\frac{1}{I(\theta)}}_{MSE\ of\ \hat{\theta}_n}.$$

(*The inequality is also known as Cramer Rao's bound)

Remarks

- The above 4 results confirms that maximum likelihood method is one of the best methods for parameter estimation.
- The result (3) is also useful for finding the confidence interval of θ.

To derive those 4 properties, we first list out the regularity conditions involved:

(1) $\ln f_X(x|\theta)$ is three times differentiable with respect to θ;

(2) $\int \frac{\partial}{\partial\theta} f_X(x|\theta)dx = \frac{\partial}{\partial\theta}\int f_X(x|\theta)dx$ and $\int \frac{\partial^2}{\partial\theta^2} f_X(x|\theta)dx = \frac{\partial^2}{\partial\theta^2}\int f_X(x|\theta)dx$.

(3) There is a function $H(x)$ such that

$$\left|\frac{\partial^3}{\partial\theta^3}\ln\big(f_X(x|\theta)\big)\right| < H(x) \ \ and \ \ \underbrace{\int_{-\infty}^{\infty} H(x) f_X(x|\theta)dx}_{\mathbb{E}[H(X)]} < \infty$$

(4) $-\infty < \int f_X(x|\theta)\frac{\partial^2}{\partial\theta^2}\ln\big(f_X(x|\theta)\big)\,dx = \mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ln\big(f_X(X|\theta)\big)\right] < 0.$

We also need the Central Limit theorem and strong law of large number:

- **Central Limit theorem** – If $X_1, X_2, \ldots$ are i.i.d. with $\mathbb{E}[X_i] = \mu < \infty$ and $Var(X_i) = \sigma^2 < \infty$, then $\frac{X_1+X_2+\cdots+X_n}{n}$ is approximately normal with mean μ and variance $\frac{\sigma^2}{n}$ when $n \to \infty$.
- **Strong law of large number** -- If $X_1, X_2, \ldots$ are i.i.d. with $\mathbb{E}[X_i] = \mu < \infty$, then $\frac{X_1+X_2+\cdots+X_n}{n} \to \mathbb{E}[X_i] = \mu$ almost surely when $n \to \infty$.

Proof of properties

Since property 1 can be deduced from property 3, it suffices to prove the properties 2, 3 and 4. Here, we shall present a rough derivation (not 100% formal) of these properties.

To facilitate the analysis, we shall let θ_{true} denotes the true value of θ.

Proof property 2

We shall argue that the MLE $\hat{\theta}_n$ will get close to θ_{true} when $n \to \infty$.

Recall that MLE $\hat{\theta}_n$ maximizes the loglikelihood function $l(\theta) = \sum_{i=1}^{n} \ln f_X(X_i|\theta)$ and hence it also maximizes $\frac{1}{n} l(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ln f_X(X_i|\theta)$.

Using strong law of large number, we deduce that

$$\frac{1}{n} l(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ln f_X(X_i|\theta) \overset{n\to\infty}{\rightarrow} \mathbb{E}[\ln f_X(X|\theta)].$$

So when $n \to \infty$, $\hat{\theta}_n$ "maximizes" $\mathbb{E}[\ln f_X(X|\theta)]$ too.

Next, we would like to argue that θ_{true} also maximizes $\mathbb{E}[\ln f_X(X|\theta)]$.

To do so, we argue that for any $\theta \neq \theta_{true}$, we have

$$\mathbb{E}[\ln f(X|\theta)] < \mathbb{E}[\ln f(X|\theta_{true})]$$

So θ_{true} is the unique maximizer of $\mathbb{E}[\ln f(X|\theta)]$.

To see this, we consider the difference

$$\mathbb{E}[\ln f(X|\theta)] - \mathbb{E}[\ln f(X|\theta_{true})] = \mathbb{E}\left[\ln\left(\frac{f(X|\theta)}{f(X|\theta_{true})}\right)\right].$$

By some calculus, one can establish that $\ln t < t - 1$ for all $t > 0$ and $t \neq 1$. Also note that $f_X(x|\theta) \neq f_X(x|\theta_{true})$ for some x when $\theta \neq \theta_{true}$, the above difference can be expressed as

$$\mathbb{E}[\ln f(X|\theta)] - \mathbb{E}[\ln f(X|\theta_{true})] = \mathbb{E}\left[\ln\left(\frac{f(X|\theta)}{f(X|\theta_{true})}\right)\right] < \mathbb{E}\left[\frac{f(X|\theta)}{f(X|\theta_{true})} - 1\right]$$

$$= \int_{-\infty}^{\infty}\left(\frac{f(x|\theta)}{f(x|\theta_{true})} - 1\right)\underbrace{f(x|\theta_{true})}_{Not\ f(x|\theta)!}dx = \underbrace{\int_{-\infty}^{\infty} f(x|\theta)dx}_{=1} - \underbrace{\int_{-\infty}^{\infty} f(x|\theta_{true})dx}_{=1} = 0$$

This shows that $\mathbb{E}[\ln f(X|\theta)] < \mathbb{E}[\ln f(X|\theta_{true})]$.

In summary, we have established that

- $\hat{\theta}_n$ maximizes the function $\frac{1}{n} l(\theta)$ and this function converges to $\mathbb{E}[\ln f(X|\theta)]$ as $n \to \infty$.
- θ_{true} maximizes the function $\mathbb{E}[\ln f(X|\theta)]$ and θ_{true} is the unique maximizer.

Provided that $f(X|\theta)$ is sufficiently smooth in θ, we expect that $\hat{\theta}_n$ should converges to θ_{true} as $n \to \infty$.

Proof of property 3

To prove this property, we first obtain an explicit formula for $\hat{\theta}_n$ if possible.

Recall that the MLE $\hat{\theta}_n$ satisfies the first order condition $l'(\hat{\theta}_n) = 0$, we apply Taylor theorem on $l'(\theta)$ (up to second order theam) and obtain

$$l'(\theta) = l'(\theta_{true}) + l''(\theta_{true})(\theta - \theta_{true}) + \frac{l'''(c)}{2!}(\theta - \theta_{true})^2.$$

where $c \in (\theta, \theta_{true})$.

By taking $\theta = \hat{\theta}_n$, we deduce that

$$0 = l'(\hat{\theta}_n) = l'(\theta_{true}) + l''(\theta_{true})(\hat{\theta}_n - \theta_{true}) + \frac{l'''(c)}{2!}(\hat{\theta}_n - \theta_{true})^2$$

$$\Rightarrow \hat{\theta}_n = \theta_{true} - \frac{l'(\theta_{true}) + \frac{l'''(c)}{2!}(\hat{\theta}_n - \theta_{true})^2}{l''(\theta_{true})}.$$

Next, we note the followings:

- $\hat{\theta}_n$ is close to θ_{true} as $n \to \infty$ (as shown in property 2). By strong law of large number and regularity condition (3), we deduce that

$$\frac{1}{n} l'''(c) \overset{l'''(\theta)=\sum \frac{\partial^3}{\partial \theta^3} \ln f(X_i|\theta)}{\rightarrow} \mathbb{E}\left[\frac{\partial^3}{\partial \theta^3} \ln f(X|c)\right] \leq \mathbb{E}[H(X)]$$

$$= \int H(x) f(x|\theta_{true}) dx < \infty.$$

We deduce that $\frac{1}{n} l'''(c)(\hat{\theta}_n - \theta_{true})^2 \overset{n\to\infty}{\rightarrow} 0$ (or $\frac{l'''(c)}{n}(\hat{\theta}_n - \theta_{true})^2 \approx 0$)

- Applying law of large number again, we can deduce that

$$\frac{1}{n} l''(\theta_{true}) \overset{l''(\theta)=\sum \frac{\partial^2}{\partial \theta^2} \ln f(X_i|\theta)}{\cong} \mathbb{E}\left[\frac{\partial^2}{\partial \theta^2} \ln f(X|\theta)\,|_{\theta=\theta_{true}}\right] = -\frac{I(\theta_{true})}{n}.$$

- Applying central limit theorem, we can show that the random variable

$$\frac{l'(\theta_{true})}{n} = \frac{\sum_{i=1}^{n} \frac{\partial}{\partial \theta} \ln f(X_i|\theta)\,|_{\theta=\theta_{true}}}{n}$$

has approximately normal distribution with mean $\mathbb{E}\left[\frac{\partial}{\partial \theta} \ln f(X|\theta)\,|_{\theta=\theta_{true}}\right]$ and variance $\frac{Var\left[\frac{\partial}{\partial \theta} \ln f(X|\theta)|_{\theta=\theta_{true}}\right]}{n}$.

It remains to compute $\mathbb{E}\left[\frac{\partial}{\partial \theta} \ln f(X|\theta)\,|_{\theta=\theta_{true}}\right]$ and $Var\left[\frac{\partial}{\partial \theta} \ln f(X|\theta)\,|_{\theta=\theta_{true}}\right]$

By direct computation and the regularity condition (2), we get

$$\mathbb{E}\left[\frac{\partial}{\partial \theta} \ln f(X|\theta)\,|_{\theta=\theta_{true}}\right] = \mathbb{E}\left[\frac{\frac{\partial}{\partial \theta} f(X|\theta_{true})}{f(X|\theta_{true})}\right] = \int_{-\infty}^{\infty} \frac{\frac{\partial}{\partial \theta} f(x|\theta_{true})}{f(x|\theta_{true})} f(x|\theta_{true}) dx$$

$$= \int_{-\infty}^{\infty} \frac{\partial}{\partial \theta} f(x|\theta_{true}) dx \overset{By\ (2)}{\cong} \frac{\partial}{\partial \theta} \underbrace{\int_{-\infty}^{\infty} f(x|\theta_{true}) dx}_{=1} = 0$$

To compute the variance, we recall that the fact that $\mathbb{E}\left[\frac{\partial}{\partial\theta}\ln f(X|\theta)\,|_{\theta=\theta_{true}}\right]=0$ so

$$Var\left[\frac{\partial}{\partial\theta}\ln f(X|\theta)\,|_{\theta=\theta_0}\right]\overset{Var(Y)=\mathbb{E}[Y^2]-(\mathbb{E}[Y])^2}{=}\mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\ln f(X|\theta)\,|_{\theta=\theta_0}\right)^2\right]$$

To compute the second moments, we consider the fact $\int_{-\infty}^{\infty}f_X(x|\theta)dx=1$

By taking the second derivative on both sides, we have

$$\frac{\partial^2}{\partial\theta^2}\int_{-\infty}^{\infty}f_X(x|\theta)dx=0\Rightarrow\int_{-\infty}^{\infty}\frac{\partial^2}{\partial\theta^2}f_X(x|\theta)dx=0\ldots(*)$$

Using the fact that $\frac{\partial}{\partial\theta}\ln f_X(x|\theta)=\frac{\frac{\partial}{\partial\theta}f_X(x|\theta)}{f_X(x|\theta)}$, we get

$$\begin{aligned}\frac{\partial^2}{\partial\theta^2}f_X(x|\theta)&=\frac{\partial}{\partial\theta}\left(f_X(x|\theta)\frac{\partial}{\partial\theta}\ln f_X(x|\theta)\right)\\&=\frac{\partial}{\partial\theta}f_X(x|\theta)\frac{\partial}{\partial\theta}\ln f_X(x|\theta)+f_X(x|\theta)\frac{\partial^2}{\partial\theta^2}\ln f_X(x|\theta)\\&=f_X(x|\theta)\left(\frac{\partial}{\partial\theta}\ln f_X(x|\theta)\right)^2+f_X(x|\theta)\frac{\partial^2}{\partial\theta^2}\ln f_X(x|\theta)\end{aligned}$$

Substitute the results into equation $(*)$, we get

$$\underbrace{\int_{-\infty}^{\infty} f_X(x|\theta)\left(\frac{\partial}{\partial\theta}\ln f_X(x|\theta)\right)^2 dx}_{=\mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\ln f_X(x|\theta)\right)^2\right]} + \underbrace{\int_{-\infty}^{\infty} f_X(x|\theta)\frac{\partial^2}{\partial\theta^2}\ln f_X(x|\theta)\,dx}_{=\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ln f_X(x|\theta)\right]} = 0$$

Combining the results, we deduce that

$$Var\left[\frac{\partial}{\partial\theta}\ln f(X|\theta)\,|_{\theta=\theta_{true}}\right] = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\ln f(X|\theta)\,|_{\theta=\theta_{true}}\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\ln f_X(x|\theta_{true})\right] = \frac{I(\theta_{true})}{n}.$$

So we deduce that $\frac{l'(\theta_{true})}{n} \sim N\left(0,\ \frac{I(\theta_{true})}{n^2}\right)$.

Combining the result, we get

$$\hat{\theta}_n = \theta_{true} - \frac{l'(\theta_{true}) + \frac{l'''(c)}{2!}\left(\hat{\theta}_n - \theta_{true}\right)^2}{l''(\theta_{true})} \approx \theta_{true} - \frac{\frac{l'(\theta_{true})}{n}}{\frac{l''(\theta_{true})}{n}} \sim \theta_{true} - \frac{\frac{l'(\theta_{true})}{n}}{-\frac{I(\theta_{true})}{n}}$$

$$\sim N\left(\theta_{true}, \frac{1}{\left(\frac{I(\theta_{true})}{n}\right)^2}\frac{I(\theta_{true})}{n^2}\right) = N\left(\theta_{true}, \frac{1}{I(\theta_{true})}\right)$$

Proof of Property 4

We let $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ be any unbiased estimator for θ such that $\mathbb{E}[\hat{\theta}|\theta] = \theta$.

To facilitate the analysis, we define

$$Z(X_1, X_2, \dots, X_n) = \frac{\partial}{\partial\theta} l(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \ln f_X(X_i|\theta).$$

Next, we consider the covariance $cov(Z, \hat{\theta}) = \rho_{Z\hat{\theta}} \sqrt{VaR(Z)} \sqrt{Var(\hat{\theta})}$. Using the fact that $-1 \leq \rho_{Z\hat{\theta}} \leq 1$, we deduce that

$$[cov(Z, \hat{\theta})]^2 \leq Var(Z) Var(\hat{\theta}) \Rightarrow Var(\hat{\theta}) \geq \frac{[cov(Z, \hat{\theta})]^2}{Var(Z)} \dots\dots (*)$$

It remains to calculate the value of R.H.S.

Recall from the proof of property 3 that $Var\left[\frac{\partial}{\partial\theta} \ln f_X(X_i|\theta)\right] = \frac{I(\theta)}{n}$, so we have

$$Var(Z) = \sum_{i=1}^{n} Var\left(\frac{\partial}{\partial\theta} \ln f_X(X_i|\theta)\right) = n\left(\frac{I(\theta)}{n}\right) = I(\theta) \dots\dots (**)$$

To compute $cov(Z, \hat{\theta})$, we recall that the fact that $\hat{\theta}$ is unbiased and consider

$$\theta = \mathbb{E}[\hat{\theta}|\theta] = \int_{\mathbb{R}^n} \hat{\theta}(x_1, x_2, \dots, x_n) \underbrace{f_X(x_1|\theta) f_X(x_2|\theta) \dots f_X(x_n|\theta)}_{f_{\vec{X}}(x_1, x_2, \dots, x_n)} dx_1 dx_2 \dots dx_n$$

Differentiating both sides of the equation with respect to θ, we have

$$1 = \int_{\mathbb{R}^n} \hat{\theta}(x_1, x_2, \dots, x_n) \left[\left(\underbrace{\frac{\partial f_X(x_1|\theta)}{\partial \theta}}_{= f_X(x_1|\theta) \frac{\partial}{\partial \theta} \ln f_X(x_1|\theta)} f_X(x_2|\theta) \dots f_X(x_n|\theta) \right) + \left(f_X(x_1|\theta) \underbrace{\frac{\partial f_X(x_2|\theta)}{\partial \theta}}_{f_X(x_2|\theta) \frac{\partial}{\partial \theta} \ln f_X(x_2|\theta)} \dots f_X(x_n|\theta) \right) \right.$$
$$\left. + \dots + \left(f_X(x_1|\theta) f_X(x_2|\theta) \dots \underbrace{\frac{\partial f_X(x_n|\theta)}{\partial \theta}}_{f_X(x_n|\theta) \frac{\partial}{\partial \theta} \ln f_X(x_n|\theta)} \right) \right] dx_1 dx_2 \dots dx_n$$

$$\Rightarrow 1 = \int_{\mathbb{R}^n} \hat{\theta}(x_1, \dots, x_n) Z(x_1, \dots, x_n) f_X(x_1|\theta) f_X(x_2|\theta) \dots f_X(x_n|\theta) dx_1 dx_2 \dots dx_n = \mathbb{E}[\hat{\theta} Z].$$

We deduce that $cov(\hat{\theta}, Z) = \underbrace{\mathbb{E}[\hat{\theta} Z]}_{=1} - \mathbb{E}[\hat{\theta}] \underbrace{\mathbb{E}[Z]}_{=0} = 1$ and hence

$$Var(\hat{\theta}) \geq \frac{[cov(Z, \hat{\theta})]^2}{Var(Z)} = \frac{1}{I(\theta)}.$$

Example 15

A random variable X has exponential distribution with p.d.f. $f_X(x|\lambda) = \lambda e^{-\lambda x},\ x > 0$. Here, $\lambda > 0$ is an unknown parameter.

A random sample $\{X_1, X_2, \dots, X_n\}$ is taken from the distribution. We let $\hat{\lambda}_n$ be the MLE. Calculate the MSE of $\hat{\lambda}_n$ when $n \to \infty$.

☺Solution

According to the property 1, $\hat{\lambda}_n$ is asymptotically unbiased so that $Bia(\hat{\lambda}_n) \to 0$ when $n \to \infty$. On the other hand, we note that

$$\mathbb{E}\left[\frac{\partial^2}{\partial\lambda^2}\ln f_X(X|\lambda)\right] = \mathbb{E}\left[\frac{\partial^2}{\partial\lambda^2}(\ln\lambda - \lambda X)\right] = \mathbb{E}\left[-\frac{1}{\lambda^2}\right] = -\frac{1}{\lambda^2}.$$

So that

$$Var(\hat{\lambda}_n) = \frac{1}{I(\theta)} = \frac{1}{-n\mathbb{E}\left[\frac{\partial^2}{\partial\lambda^2}\ln f_X(X|\lambda)\right]} = \frac{\lambda^2}{n}.$$

Hence, $MSE(\hat{\lambda}_n) = Var(\hat{\lambda}_n) + \left[Bia(\hat{\lambda}_n)\right]^2 \approx \frac{\lambda^2}{n}$.

What is the distribution of MLEs for multiple parameters? (Harder)

In fact, one can extend the above result to the case when there are multiple parameters (applicable to normal distribution, pareto distribution etc.)

We let $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k$ be the MLEs of the parameters $\theta_1, \theta_2, \dots, \theta_k$ in a probability distribution function. Provided that the sample size n is sufficiently large, $\vec{\hat{\theta}} = \left(\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k\right)$ has multivariate normal distribution with mean $\vec{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$ and covariance matrix $\Sigma = \left(I(\theta)\right)^{-1}$ (i.e. $\hat{\theta} \sim N_k\left(\vec{\theta}, \Sigma\right)$), where

$$I\left(\vec{\theta}\right) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix},$$

where

$$a_{ij} = \mathbb{E}\left[\frac{\partial}{\partial\theta_i \partial\theta_j} l\left(\vec{\theta}\right)\right] = -n\mathbb{E}\left[\frac{\partial}{\partial\theta_i \partial\theta_j} \ln f\left(X\middle|\vec{\theta}\right)\right].$$

Model fitting and empirical distribution

So far, we have developed a methodology for constructing a parametric model for the random variable X. One problem of this approach is that we assign a specific probability distribution on the population (e.g. normal distribution) and do "minor tuning" by adjusting the parameters (e.g. μ, σ in normal distribution). However, the resulting model can be very different from the correct distribution if our initial guess is incorrect.

Hence, one needs to check if the proposed model is consistent to the actual distribution of sample data. To do so, one needs to "estimate" the true distribution and this can be done using *histogram* or *empirical distribution*.

Given a random sample $\{X_1, X_2, \dots, X_n\}$ drawn from the distribution of X, we estimate the probability distribution of X (i.e. $F_X(x) = P(X \le x)$) by

$$F_n(x) = \frac{Number\ of\ data \le x}{n} = \frac{\sum_{i=1}^{n} \mathbf{1}_{\{X_i \le x\}}}{n},$$

where $\mathbf{1}_{\{A\}} = \begin{cases} 1 & if\ \ A\ happens \\ 0 & if\ \ A\ does\ not\ happen \end{cases}$ denotes *indicator function.*

The estimate $F_n(x)$ is called empirical distribution, which is used to approximate the true distribution of X.

Justification of the approximation method

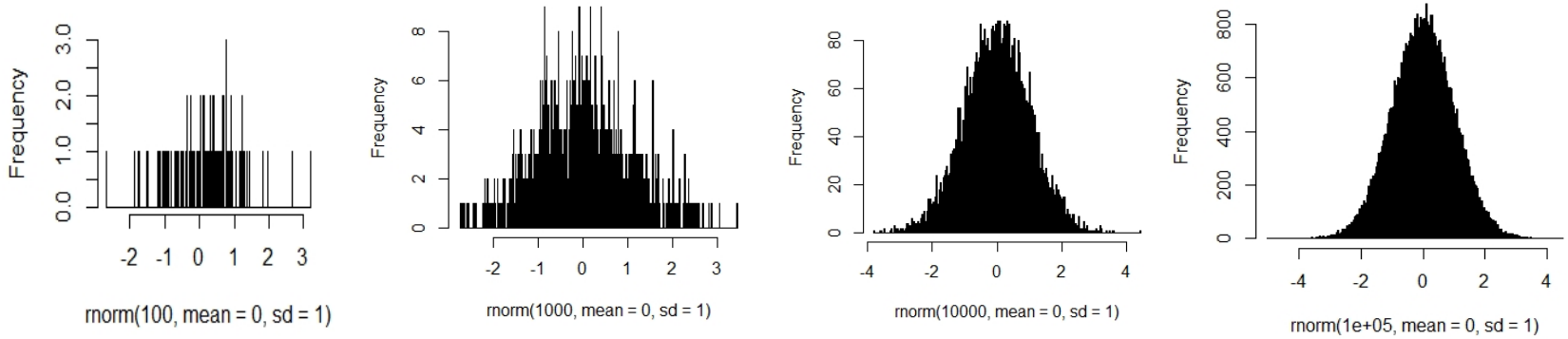
We let $Y_i = \mathbf{1}_{\{X_i \le x\}}$. Since $Y_1, Y_2, Y_3, \ldots$ are independent and identically distributed and $\mathbb{E}[|Y_i|] = \mathbb{E}\left[\mathbf{1}_{\{X_i \le x\}}\right] \le 1 < \infty$, it follows from strong law of large number that

$$F_n(x) = \frac{\sum_{i=1}^{n} \mathbf{1}_{\{X_i \le x\}}}{n} \overset{n\to\infty}{\rightarrow} \mathbb{E}\left[\mathbf{1}_{\{X_i \le x\}}\right] = P(X_i \le x) = F_X(x).$$

almost surely when $n \to \infty$. Theoretically, the empirical distribution can approximate the true distribution (i.e. $F_X(x)$) when n is sufficient large.

As a numerical example, we consider a random sample drawn from standard normal distribution and examine the empirical distribution:

$n = 100$ $n = 1000$ $n = 10000$ $n = 100000$



We observe that the empirical distribution converges to true distribution when $n \to \infty$.

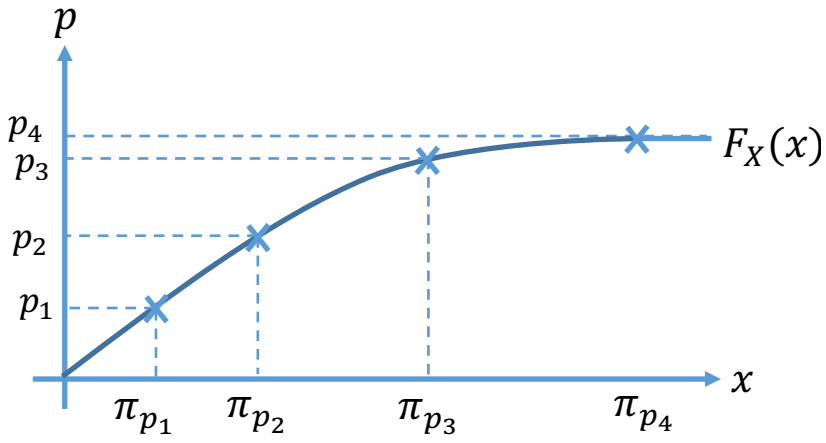
Percentile and QQ-plot

We let X be a random variable with certain probability distributions (represented by $F_X(x) = P(X \leq x)$. For any $p \in (0,1)$, we define the $100p$-th percentile of X as a number π_p satisfying

$$F_X(\pi_p) = p.$$

- If X is a continuous random variable which takes values over an interval (bounded or unbounded), then $F_X(\pi_p) = p$ will have unique solution.
- If X is discrete random variable, $F_X(\pi_p) = p$ may not have solution. In this case, we define the $100p$-th percentile of X as $\pi_P = \sup\{x|F_X(x) < p\} \quad or \quad \inf\{x|F_X(x) \geq p\}$.

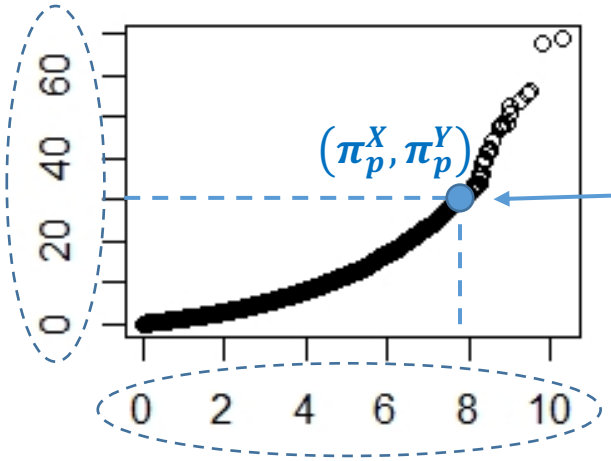
Graphically, each percentile represents a point on $y = F_X(x)$ in x-p plane. Thus, the set of percentiles $\{\pi_p : 0 < p < 1\}$ should determine the probability distribution of a random variable X.



Based on the above observation, one can deduce the following conjecture

$$\left\{\begin{matrix} X \text{ and } Y \text{ has same} \\ \text{distribution } F_X(z) = F_Y(z) \end{matrix}\right\} \Leftrightarrow \left\{\begin{matrix} \pi_p^X = \pi_p^Y \\ \text{for all } p \in (0,1) \end{matrix}\right\}.$$

Given this fact, one can compare the distributions of X and Y using QQ-plot. Roughly speaking, QQ-plot is a plot of percentiles of one distribution against the percentiles of another distribution. Here's an example of QQ-plot:



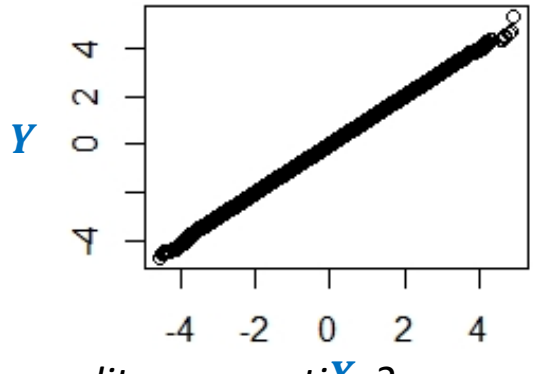
percentiles of Y (from $p = 0$ to $p =$

Each point represents the percentiles of X and Y for a given value of p

percentiles of X (from $p = 0$ to $p = 1$)

(*Note: In R, one can obtain the qq-plot using the command "qqplot(x,y)")

If X and Y have same distribution, then they have same set of percentiles (i.e. $\pi_p^X = \pi_p^Y$) and QQ-plot will show a straight line with slope 1.



How do we use QQ-plot to check normality assumption?

Suppose that a random variable X follows normal distribution with mean μ and standard deviation σ, then X can be expressed as

$$X = \mu + \sigma Z,$$

where Z is standard normal random variable.

We let π_p^Z be the 100p-th percentile of Z. Then the corresponding 100p-th percentile of X can be expressed as

$$F_X(\pi_P^X) = p \Rightarrow P(\mu + \sigma Z \leq \pi_P^X) = p \Rightarrow P\left(Z \leq \frac{\pi_P^X - \mu}{\sigma}\right) = p \Rightarrow \frac{\pi_P^X - \mu}{\sigma} = \pi_P^Z \Rightarrow \pi_P^X$$
$$= \mu + \sigma\pi_P^Z.$$

Based on the above result, one can check the normality of X (or sample data) by plotting a QQ-plot of X against the standard normal distribution Z

- If the QQ-plot shows a (roughly) straight line (see Figure 1), then X is likely to be normally distributed.
- If the QQ-plot does not form a straight line (see Figure 2), then X may not be normally distributed.

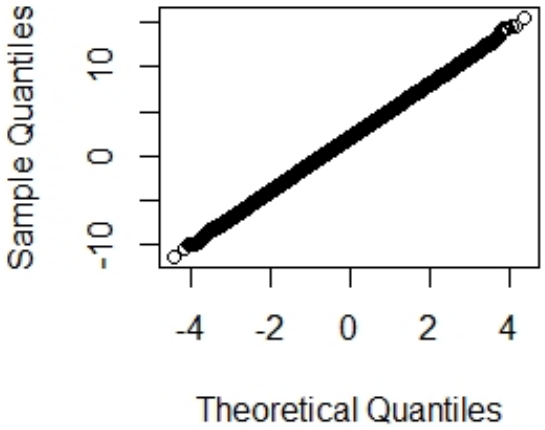


Figure 1: Sample data is drawn from normal distribution

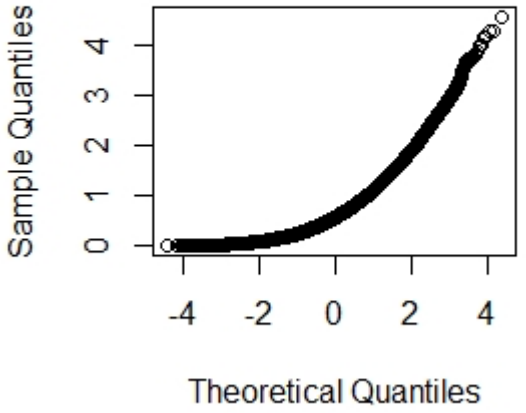


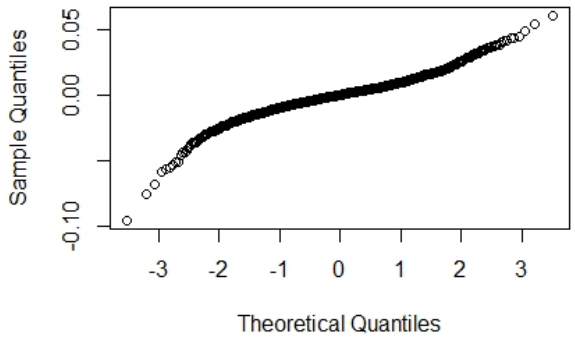
Figure 2: Sample data is drawn from gamma distribution

Example 16

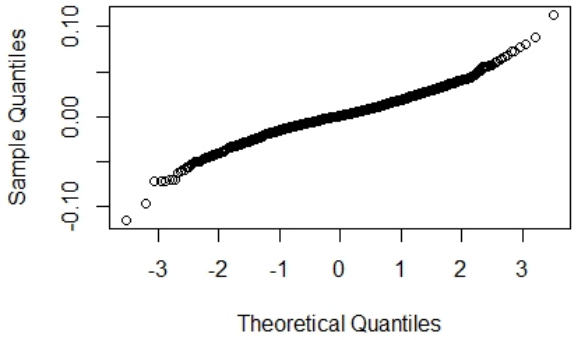
Let us revisit empirical distribution of two stocks tradable in Hong Kong market (Code: 0005.HK and 0700.HK) considered in Example 14 and check whether the distributions of stock returns are normally distributed using QQ-plots.

(*Note: x-axis represents the percentiles of standard normal)

0005.HK



0700.HK



It appears that the both QQ-plots do not form a straight line. Since the empirical distribution can approximate the true distribution, it follows that normality assumption may not hold.

Hypothesis Testing on normality check

Although QQ-plot can provide a quick scan on the validity of the normality assumption, it may not able to draw an accurate conclusion since the decision is often made based on subjective judgment.. In order to get a more precise prediction, we will examine this using hypothesis testing.

There are a lot of different test available for this purpose. In this section, we will explore the most intuitive test so called *Kolmogorov-Smirnov test*.

Principle of Kolmogorov-Smirnov test (KS-test)

This test aims to test whether the sample data comes from a certain probability distribution (denoted by $F_X(x)$).

Under this test, the two hypothesis will be set as follows:

H_0: $The\ sample\ data\ is\ drawn\ from\ the\ probability\ distribution\ F_X(x)$

H_1: $The\ sample\ data\ is\ NOT\ drawn\ from\ the\ probability\ distribution\ F_X(x)$

(*In our case, this distribution $F_X(x)$ is the normal distribution)

What is the test statistic?

Given the sample data $\{X_1, X_2, \dots, X_n\}$, we let $F_n(x)$ be the empirical distribution of Z obtained from this sample data.

When n is sufficiently large, $F_n(x)$ can roughly approximate the true distribution. So one can check the normality by comparing the empirical distribution with the normal distribution (denoted by $F_X(x)$) and see whether there is any significant difference.

Therefore, we consider the following statistics:

$$D_n = \sup_{x \in (-\infty, \infty)} |F_n(x) - F(x)| \geq 0.$$

If X_i is normally distributed, then D_n should be very close to 0 especially when $n \to \infty$. Thus, we deduce that

$$\left\{ \begin{matrix} Reject\ H_0\ \ and \\ Accept\ H_1 \end{matrix} \right\} \Leftrightarrow D_n > D^*,$$

where D^* is a critical threshold.

Example 17 (Check the normality of distribution of daily return)

We consider the daily log-return of a stock "0005.HK" and "0700.HK" over the period 1st Jan, 2011- 31st Dec, 2019. It would like to examine whether the return is normally distributed at 1% significant level.

For 0005.HK

```
library(MASS)
X0005_HK<-na.omit(X0005_HK)
x<-X0005_HK$Logreturn
x<-x+0.0000000000000001*rnorm(length(x),0,1) #This is for avoiding ties in the dataset

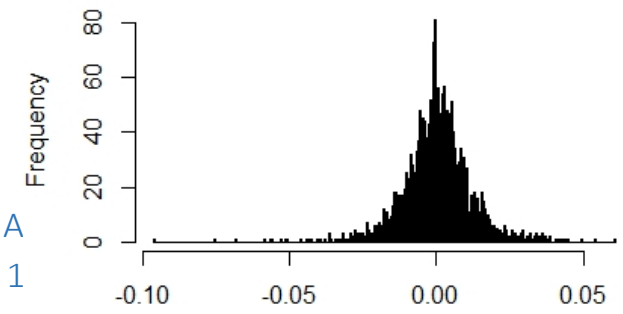
MLEs<-fitdistr(x,"normal") #Calcualte the MLE of the parameters

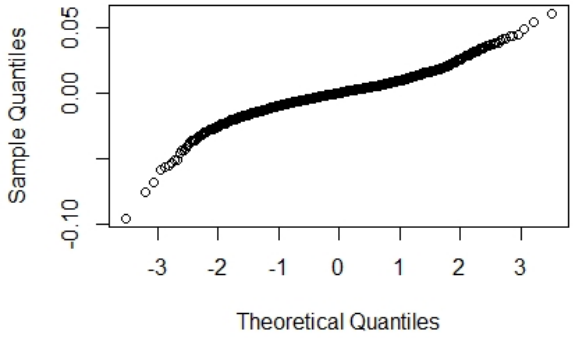
ks.test(x,'pnorm',mean=MLEs$estimate["mean"],sd=MLEs$estimate["sd"]) #Perform ks.test

        Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.06192, p-value = 8.535e-08
alternative hypothesis: two-sided
```

Since the p-value $< 0.01 = \alpha$, so one can reject H_0 and conclude that the daily return of 0005.HK is *not* normally distributed.





For 0700.HK

```
library(MASS)
X0700_HK<-na.omit(X0700_HK)
x<-X0700_HK$Logreturn
x<-x+0.00000000000000001*rnorm(length(x),0,1) #This is for avoiding ties in the d
ataset

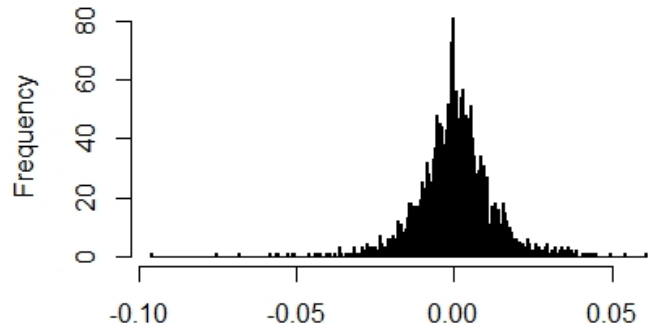
MLEs<-fitdistr(x,"normal") #Calcualte the MLE of the parameters

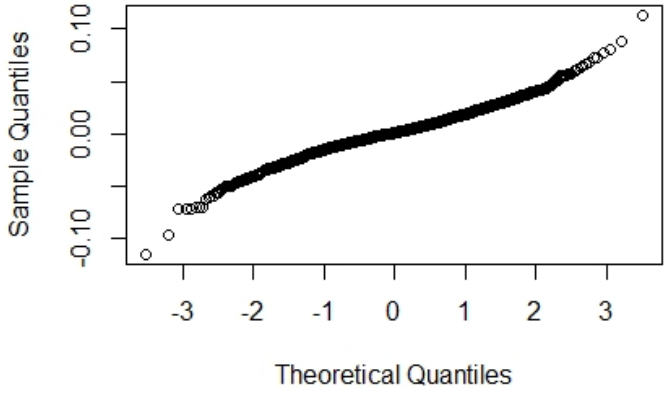
ks.test(x,'pnorm',mean=MLEs$estimate["mean"],sd=MLEs$estimate["sd"]) #Perform k
s.test

      Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.047371, p-value = 9.719e-05
alternative hypothesis: two-sided
```

Since the p-value $< 0.01 = \alpha$, so one can accept H_1 and conclude that the daily return of 0700.HK is *not* normally distributed.





Remark of Example 17 (Better distribution for stock return?)

We observe from the above example that normal distribution may not be a suitable candidate for modelling the stock return.

Instead, one may consider general t-distribution which is defined as follows:

- We let $T_\nu = \frac{Z}{\sqrt{\chi^2_\nu/\nu}}$ be student-t distribution with degree of freedom $\nu > 0$. The random variable defined by
$$X = \mu + \lambda T_\nu$$
is said to have general t-distribution with degree of freedom ν and parameters μ (location parameter) and λ (scale parameter). We denote it by $X \sim t_\nu(\mu, \lambda)$.
- One can show that $\mathbb{E}[X] = \mu$ and $Var(X) = \frac{\nu}{\nu-2}\lambda^2$ for $\nu > 2$.
- On the other hand, the parameter ν can indicate the heaviness of the tail distribution. In particular, the distribution has heavier tail when ν takes smaller value.

Next, we perform KS-test again and see whether *t*-distribution can fit the data well. In R, one can perform KS-test (for t-distribution) through the following command:

```
library("Rfast")
library("ramify")
library("LambertW")

x<-X0005_HK$Logreturn #Store the target data
x<-na.omit(x)           #remove "NA" in the dataset
x<-as.matrix(x)         #ensure that the data is in matrix form
df=seq(2,8,0.001)       #set the range of degree of freedom (e.g.
n=length(df)
loglik=rep(0,n)         #Used to store the value of loglikelihood
function for different level of df

#For each df, calculate the MLE and loglikelihood function
for(i in 1:n){
  fitresult=mvt.mle(x,v=df[i])
  loglik[i]=fitresult$loglik
}

dfmax=df[which.max(loglik)] #Find the maximizier(optimal df)
fitMLE=mvt.mle(x,v=dfmax) #Estimate the MLE of the parameters
m<-fitMLE$location             #m=mu (location)
s<-sqrt(fitMLE$scatter[1,1])  #s=lambda (scale parameter)


T=(x-m)/s
ks.test(T,'pt',df=dfmax) #do the testing using ks.test
```

Output result (0005.HK)

```
> print(m)
[1] 0.0002001957
> print(s)
[1] 0.008415141
> print(dfmax)
[1] 3.631
> ks.test(T,'pt',df=dfmax) #do the testing using ks.test

    Asymptotic one-sample Kolmogorov-Smirnov test

data:  T
D = 0.019484, p-value = 0.3702
alternative hypothesis: two-sided
```

Output result (0700.HK)

```
> print(m)
[1] 0.001117649
> print(s)
[1] 0.01526518
> print(dfmax)
[1] 4.937
> ks.test(T,'pt',df=dfmax) #do the testing using ks.test

    Asymptotic one-sample Kolmogorov-Smirnov test

data:  T
D = 0.01842, p-value = 0.4406
alternative hypothesis: two-sided
```

We observe that the p-values of two stock are quite high. So we accept H_0 in the case. It appears that t-distribution may be a better candidate for modeling stock return (*Note: However, it does not imply that the stock return must follow t-distribution due to the nature of hypothesis testing.

Remarks about KS-test – Technical Aspect

In this section, we present a method for calculating the test statistic for KS-test manually. Recall that the test statistics of KS-test is given by

$$D = \sup_{x\in(-\infty,\infty)} |F_n(x) - F_X(x)|$$

We let $\{X_1, X_2, \dots, X_n\}$ be the sample data and the data is sorted such that $X_1 \leq X_2 \leq X_3 \leq X_4 \leq \cdots \leq X_n$.

Note that $F_n(x)$ is constant over the interval $[X_{i-1}, X_i)$ for $i = 1,2, \dots, n+1$ (*Here, we take $X_0 = -\infty$ and $X_{n+1} = +\infty$) and $F_X(x)$ is increasing function.

Then one can show that

$$D = \sup_{x\in(-\infty,\infty)} |F_n(x) - F_X(x)| = \sup_{i=1,2,\dots,n+1} \left(\sup_{x\in[X_{i-1},X_i)} |F_n(x) - F_X(x)| \right)$$
$$= \sup_{i=1,2,\dots,n+1} (\max(|F_n(X_{i-1}) - F_X(X_{i-1})|, |F_n(X_{i-1}) - F_X(X_i)|))$$

(Note: We take $F_X(X_0) = \lim_{x\to-\infty} F_X(x) = 0$ and $F_X(X_{n+1}) = \lim_{x\to\infty} F_X(x) = 1$.)

As an example, we revisit the analysis of 0700.HK in Example 17 (against normal). Here is the sample code for calculating the test-statistics:

```
x<-X0700_HK$Logreturn #Store the target data
x<-na.omit(x)
#Obtain MLE of the parameters (Normal distribution)
MLEs=fitdistr(x,"normal")
mu=MLEs$estimate["mean"]
sigma=MLEs$estimate["sd"]
x<-sort(x,decreasing=FALSE) #Sort the data from the smallest to the largest
n=length(x) #Calculate the sample size of the data

#Write a function to calculuate the cdf of normal distribution
cdfnormal<-function(x,mu,sigma){
  cdf=pnorm(x,mean=mu,sd=sigma)
  return(cdf)
}

Dint=abs(cdfnormal(x[1],mu,sigma))
Dtest=rep(0,n) #Set an array to store the values of |Fn-FX| for calculuaing D
for (i in 1:n){
  Fn=i/n;
  D1test=abs(Fn-cdfnormal(x[i],mu,sigma)) #For calculating |Fn(X_i)-FX(X_i)|
  if (i<n){
    D2test=abs(Fn-cdfnormal(x[i+1],mu,sigma)) #For calculating |Fn(X_i)-FX(X_(i+1))|
  } else if (i==n){
    D2test=0
  }
  Dtest[i]=max(D1test,D2test)
}
D=max(Dint,max(Dtest))
> D
[1] 0.04737089
```

Anderson-Darling Test

This is another approach to test whether the sample data comes from certain distribution. It is effective when we want to focus the fitting of tail distribution (which the KS-test may not be useful).

Similar to KS-test, the hypothesis of the test is as follows:

$$H_0\text{: } The\ sample\ data\ is\ drawn\ from\ the\ distribution\ \ F_X(x)$$

$$H_1\text{: } The\ sample\ data\ is\ not\ drawn\ from\ the\ distribution\ \ F_X(x)$$

Given the sample data $\{X_1, X_2, \dots, X_n\}$ and the associated empirical distribution $F_n(x)$, the test checks the validity of the hypothesis by measuring the *weighted average distance between* $F_n(x)$ *and* $F_X(x)$.

One possible test statistics can be

$$D_n = \mathbb{E}\left[\left(F_n(X) - F_X(X)\right)^2\right] = \int_{-\infty}^{\infty} \left(F_n(x) - F_X(x)\right)^2 \underbrace{f_X(x)dx}_{weight}.$$

However, this indicator may not be useful for our purpose since the indicator assigns very small weight to tail distribution where $f_X(x)$ is very small.

To resolve this problem, one would modify the test indicator by adding an addition weight factor $\phi(x)$ as follows

$$D_n = \mathbb{E}\left[\left(F_n(X) - F_X(X)\right)^2 \phi(x)\right] = \int_{-\infty}^{\infty} \left(F_n(x) - F_X(x)\right)^2 \phi(x) \underbrace{f_X(x)dx}_{weight}.$$

Our goal is to choose $\phi(x)$ so that more weight can be assigned to tail distribution.

To do so, we examine the average value of $\left(F_n(x) - F_X(x)\right)^2$.

- Provided that H_0 is true, then the quantity $\{number\ of\ data\ X_i \le x\} = \sum_{i=1}^{n} \mathbf{1}_{\{X_i \le x\}}$ follows binomial distribution with parameters n and probability $p = F_X(x)$. Since $F_n(x) = \frac{\{number\ of\ data\ X_i \le x\}}{n}$, it follows that

$$\mathbb{E}[F_n(x)] = \frac{nF_X(x)}{n} = F_x(x) \ \ and$$

$$\mathbb{E}\left[\left(F_n(x) - F_X(x)\right)^2\right] = Var\left(F_n(x)\right) = \frac{nF_X(x)\left(1 - F_X(x)\right)}{n^2}$$

$$= \frac{F_X(x)\left(1 - F_X(x)\right)}{n}.$$

- Hence, one can choose the weight factor to be

$$\phi(x) = \frac{n}{F_X(x)\big(1 - F_X(x)\big)}.$$

  - ✓ Then it follows that $\mathbb{E}\left[\big(F_n(x) - F_X(x)\big)^2 \phi(x)\right] = 1$. This reduces the *bias* when calculating the average distance.
  - ✓ Note that the function $u(1-u) = u - u^2$ attains the minimum at $u = 0$ and $u = 1$ respectively. It follows that $\phi(x)$ will attain maximum when $F_X(x)$ is close to 0 or 1 (which corresponds to the tail part of the distribution). So it implicitly assign a greater weight to tail distribution.

Combining the above results, the test statistics of Anderson-Darling test is found to be

$$D_n^* = n \int_{-\infty}^{\infty} \frac{\big(F_n(x) - F_X(x)\big)^2}{F_X(x)\big(1 - F_X(x)\big)} f_X(x) dx.$$

Naturally, one would reject H_0 if D_n^* is sufficient large in the sense that $D_n^* \geq D^*$.

Working with Anderson-Darling test

One technical problem is that it is tedious to compute D_n^* directly since the integral is difficult to evaluate in general. Next, we would like to develop an efficient way of using this test.

- Suppose that H_0 is true and X has the probability distribution $F_X(x)$, it follows that the random variable $Y = F_X(X)$ follows uniform distribution over $[0,1]$ since for any $u \in [0,1]$,

$$P(Y = F_X(X) \leq u) = P\left(X \leq F_X^{-1}(u)\right) = F_X\left(F_X^{-1}(u)\right) = u,$$

  so the probability density function of Y is $f_Y(u) = \frac{d}{du}\underbrace{P(Y \leq u)}_{=u} = 1$.

- To check whether the sample data $\{X_1, \dots, X_n\}$ is drawn from the distribution $F_X(x)$, one can check whether the data $\{F_X(X_1), \dots, F_X(X_n)\}$ is drawn from the uniform distribution over $[0,1]$.
- Since the distribution function for uniform distribution is relatively simple (i.e. $F_U(u) = 1$ for $0 \leq u \leq 1$ in our case), so one can obtain a simple formula for D_n^*.

To facilitate the analysis, we assume that the sample data has been *sorted* such that $X_1 \leq X_2 \leq \cdots \leq X_n$. For $k = 1,2, \ldots, n$, we let $u_k = F_X(X_k) \in [0,1]$.

So the sample data becomes $\{u_1, u_2, \ldots, u_n\}$ after the transformation.

Since $F_U(u) = u$ for $0 \leq u \leq 1$ and $F_n(u) = \frac{k}{n}$ for $u_k \leq u < u_{k+1}$, it follows that

$$D_n^* = n \int_{-\infty}^{\infty} \frac{\big(F_n(u) - F_U(u)\big)^2}{F_U(u)\big(1 - F_U(u)\big)} f_U(u) du = n \int_0^1 \frac{(F_n(u) - u)^2}{u(1-u)} (1) du$$

$$= n \sum_{k=0}^{n} \left(\int_{u_k}^{u_{k+1}} \frac{\left(\frac{k}{n} - u\right)^2}{u(1-u)} du \right) = n \sum_{k=0}^{n} \left(\int_{u_k}^{u_{k+1}} \left(\frac{\left(\frac{k}{n}\right)^2}{u} + \frac{\left(\frac{k}{n}\right)^2}{1-u} - \frac{\left(\frac{2k}{n} - 1\right)}{1-u} - 1 \right) du \right)$$

$$= n \sum_{k=0}^{n} \left[\left(\frac{k}{n}\right)^2 \ln u - \left(\frac{k}{n} - 1\right)^2 \ln(1-u) - u \right]_{u_k}^{u_{k+1}}$$

$$= \cdots = -\sum_{k=1}^{n} \frac{2k-1}{n} [\ln u_k + \ln(1 - u_{n+1-k})] - n$$

$$= -\sum_{k=1}^{n} \frac{2k-1}{n} \big[\ln F_X(X_k) + \ln\big(1 - F_X(X_{n+1-k})\big)\big] - n.$$

Example 18

We revisited Example 17 again and check the normality using Anderson-Darling test. The code and output are given as follows:

For 0005.HK

```
library("goftest")
library("MASS")
x<-X0005_HK$Logreturn #Store the target data
x<-na.omit(x)

MLEs=fitdistr(x,"normal")
> ad.test(x,"pnorm",mean=MLEs$estimate['mean'],sd=MLEs$estimate['sd'])

        Anderson-Darling test of goodness-of-fit
        Null hypothesis: Normal distribution
        with parameters mean = 8.18639150474469e-05, sd = 0.0121416685993203
        Parameters assumed to be fixed

data:  x
An = 20.141, p-value = 2.711e-07
```

For 0700.HK (Skipped the coding part since it is similar to previous case)

```
> ad.test(x,"pnorm",mean=MLEs$estimate['mean'],sd=MLEs$estimate['sd'])

        Anderson-Darling test of goodness-of-fit
        Null hypothesis: Normal distribution
        with parameters mean = 0.00108974535897169, sd = 0.0195335436417075
        Parameters assumed to be fixed

data:  x
An = 10.106, p-value = 5.818e-06
```

We observe that the p-values are both less than $\alpha = 0.01$. So we can reject H_0 and conclude that the daily returns of both stocks are not normally distributed.

On the other hand, one can perform the test again t-distribution. The sample code and the output are as follows (We used the parameter values (MLEs) obtained in Example 17)

For 0005.HK

```
> print(m)
[1] 0.0002001957
> print(s)
[1] 0.008415141
> print(dfmax)
[1] 3.631
T=(x-m)/s
> ad.test(T,'pt',df=dfmax) #do the testing using ks.test

        Anderson-Darling test of goodness-of-fit
        Null hypothesis: Student's t distribution with parameter df = 3.631
        Parameters assumed to be fixed

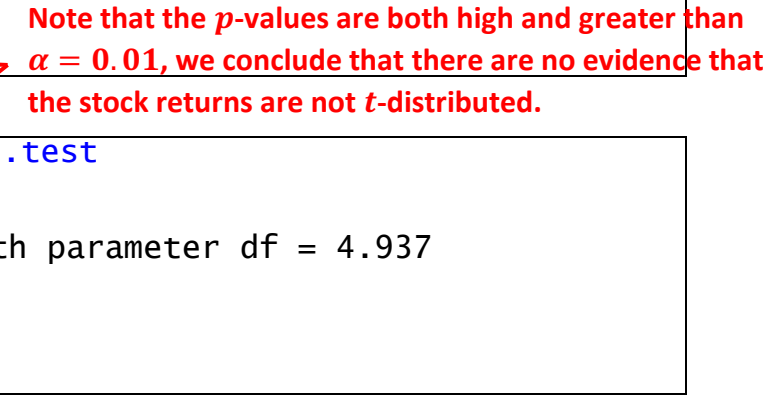
data:  T
An = 0.36482, p-value = 0.8828
```

For 0700.HK

```
> ad.test(T,'pt',df=dfmax) #do the testing using ks.test

        Anderson-Darling test of goodness-of-fit
        Null hypothesis: Student's t distribution with parameter df = 4.937
        Parameters assumed to be fixed

data:  T
An = 0.82584, p-value = 0.4628
```



Additional techniques for model selection and construction

In this section, we shall explore some advanced methods for model construction.

Bayesian Analysis (for parameter estimation)

One limitation of maximum likelihood method is that we need a large sample in order to obtain an accurate estimate for an unknown parameter. When the sample size is limited, one cannot obtain the estimate using sample data only.

Bayesian Analysis aims to estimate the unknown parameter using both sample data and other information (known as prior estimate). In addition, one can also update the estimate when there are more information/sample data available. Therefore, this method resolves the limitation of maximum likelihood method.

Problem formulation

We let X be a random variable with the probability density function $f_X(x|\theta)$, where θ is unknown parameter.

The Bayesian analysis for parameter estimation consists of two major steps:

Step 1: Initialization stage (Prior estimation)

Firstly, we consider the case when there is no sample data available and try to make a "guess" on the possible value of θ using other information.

- For example, if we wish to estimate the mean μ and variance σ^2 of return of a new stock in e-commerce industry, one can investigate the returns of other e-commerce stocks with good number of sample data. We let $(\mu_1, \sigma_1^2), (\mu_2, \sigma_2^2), \dots, (\mu_n, \sigma_n^2)$ be the estimates from these stocks. One may conjecture that (μ, σ^2) can be one of these values.

Based on these information, one can obtain a set of possible values of θ and the corresponding distribution. We denotes $\pi(\theta)$ be the distribution of θ obtained and it is called as *prior distribution*.

Remarks

In some cases, we assume $\pi(\theta)$ to be uniform distribution over a set of possible values of θ. Such prior is known as *flat prior* (used in deriving BIC).

Step 2: Updating the distribution of θ using sample data

We let $\{X_1, X_2, \dots, X_n\}$ be the sample data drawn from the distribution of X.

One can obtain the updated distribution of θ using the Bayes' rule. That is,

$$\pi(\theta|X_1, X_2, \dots, X_n) = \frac{P(X_1, X_2, \dots, X_n|\theta)}{P(X_1, X_2, \dots, X_n)} = \frac{\overbrace{f_{\vec{X}}(X_1, X_2, \dots, X_n|\theta)}^{joint\ density\ of\ X_1,\dots,X_n}\ \pi(\theta)}{\int_{-\infty}^{\infty} f_{\vec{X}}(X_1, X_2, \dots, X_n|\theta)\pi(\theta)d\theta}$$

$$= \frac{[\prod_{i=1}^{n} f_X(X_i|\theta)]\pi(\theta)}{\int_{-\infty}^{\infty}[\prod_{i=1}^{n} f_X(X_i|\theta)]\pi(\theta)d\theta}.$$

This updated distribution is known as *posterior distribution* of θ.

Remarks about Bayesian Analysis

- Different from the maximum likelihood method, Bayesian Analysis only yields a distribution of θ ("range" of θ) instead of a point estimate. However, this distribution will converge to a single estimate (true value) when $n \to \infty$.

Example 19

A random variable X has the exponential distribution with the probability density function $f_X(x|\lambda) = \lambda e^{-\lambda x},\ x > 0.$

The prior distribution of λ is given by

$$\pi(1) = \pi(2) = \pi(3) = \pi(4) = \pi(5) = 0.2.$$

We let $\{X_1, X_2, \dots, X_n\}$ be the random sample drawn from the distribution of X. Derive a general formula for posterior distribution of λ.

😊Solution

Using the general formula above, the posterior distribution of λ is seen to be

$$\pi(\lambda|X_1, \dots, X_n) = \frac{\left(\prod_{i=1}^{n} \lambda e^{-\lambda X_i}\right)\pi(\lambda)}{\sum_{k=1}^{5}\left(\prod_{i=1}^{n} k e^{-k X_i}\right)\pi(k)} \overset{\substack{\pi(k)=0.2 \\ for\ k=1,2,\dots,5}}{=} \frac{\lambda^n e^{-\lambda \sum_{i=1}^{n} X_i}}{\sum_{k=1}^{5} k^n e^{-k \sum_{i=1}^{n} X_i}}$$

for $\lambda = 1,2, \dots,5.$

Akaike Information Criterion (for model selection)

Akaike Information Criterion (AIC in short) is one of the popular methods in selecting the best probability distribution of a random variable.

Problem formulation

We let X be a random variable with the probability density function $f_X(x)$ (i.e. true distribution of X).

Given a parametric model, we let $g_X(x|\theta)$ be the probability density function of X generated from the model. Then we define

$$D = \int_{-\infty}^{\infty} (\ln f_X(x) - \ln g_X(x|\theta)) f_x(x) dx = \mathbb{E}[\ln f_X(X) - \ln g_X(X|\theta)]$$

to be the average distance between the true distribution and the proposed distribution. This distance is called Kullback-Leubler distance.

Our goal is to choose a model distribution $g_X(x|\theta)$ to minimize the distance D.

Quick remarks

Using the inequality $\ln x \leq x - 1$ for all $x > 0$ (equality hold when $x = 1$), we deduce that

$$D = \mathbb{E}[\ln f_X(X) - \ln g_X(X|\theta)] = -\mathbb{E}\left[\ln \frac{g_X(X|\theta)}{f_X(X)}\right] \geq -\mathbb{E}\left[\frac{g_X(X|\theta)}{f_X(X)} - 1\right]$$

$$= -\int_{-\infty}^{\infty} \left(\frac{g_X(x|\theta)}{f_X(x)} - 1\right) f_X(x)dx = \underbrace{\int_{-\infty}^{\infty} f_X(x)dx}_{=1} - \underbrace{\int_{-\infty}^{\infty} g_X(x|\theta)dx}_{=1} \geq 0.$$

This verifies that D is always non-negative.

Since $\mathbb{E}[\ln f_X(X)]$ is fixed, it suffices to choose $g_X(X|\theta)$ to maximize $\mathbb{E}[\ln g_X(X|\theta)]$ so that D is minimized.

How to estimate $\mathbb{E}[\ln g_X(X|\theta)]$?

One important step is to provide a value of θ. Practically, the parameter θ is estimated using the sample data $\{X_1, X_2, \ldots, X_n\}$ and we use the corresponding maximum likelihood estimate (i.e. $\hat{\theta}_n$) to estimate θ.

Assuming the sample size is large, we get

$$\mathbb{E}[\ln g_X(X|\theta)] \approx \frac{1}{n}\underbrace{\sum_{i=1}^{n} \ln g_X(X_i|\theta)}_{loglikelihood} \approx \frac{1}{n} l(\theta).$$

Given that $\hat{\theta}_n$ maximizes $l(\theta)$, we observe from the above equation that $\hat{\theta}_n$ can actually maximize the $\mathbb{E}[\ln g_X(X|\theta)]$.

Remarks

We observe from the above equation that AIC can be viewed as an extension of maximium likelihood method.

- For maximum likelihood method, we choose the parameter θ to maximize the loglikelihood function $l(\theta)$.
- For AIC, we choose both models and model parameters to maximize the loglikelihood function $l(\theta)$.

Then, we deduce that the point estimate of $\mathbb{E}\left[\ln g_X\left(X\middle|\hat{\theta}_n\right)\right]$ is $\frac{1}{n} l\left(\hat{\theta}_n\right)$.

Next, we shall check if the estimator is unbiased. When n is large, one can use the property of MLE and show that

$$\mathbb{E}\left[\frac{1}{n}l(\hat{\theta}_n)\right] = \cdots \approx \mathbb{E}\left[\ln g_X(X|\hat{\theta}_n)\right] + \frac{k}{n}.$$

Here, k denotes the number of parameters in the parametric model $g_X(x|\theta)$

This implies that the unbiased estimator of $\mathbb{E}\left[\ln g_X\left(X|\hat{\theta}_n\right)\right]$ should be

$$\frac{1}{n}l(\hat{\theta}_n) - \frac{k}{n} = -\frac{1}{2n}\underbrace{\left(-2l(\hat{\theta}_n) + 2k\right)}_{AIC\ score}.$$

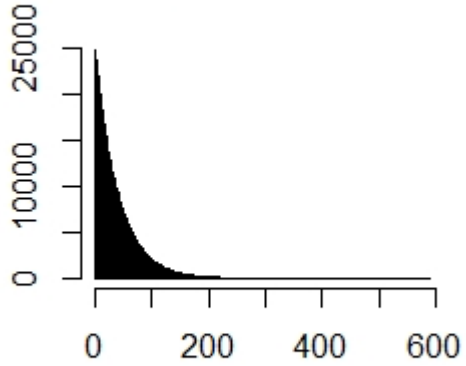
Hence, we can define the AIC score as

$$AIC = -2\begin{pmatrix} maximum \\ loglikelihood \end{pmatrix} + 2k.$$

Then the best model is the one with the smallest AIC score (so $\mathbb{E}\left[\ln g_X\left(X|\hat{\theta}_n\right)\right]$ or D could be minimized.)

Example 20 (AIC on model selection I, Test example)

We consider a random sample of size 1000000 drawn from exponential distribution with $\lambda = 0.025$. The histogram of the data is shown below:



Suppose that we wish to fit the data by one of the following distributions:

(1) exponential distribution, (2) normal distribution and (3) lognormal distribution.

Determine the best distribution using AIC criterion.

😊Solution

Recall that the Akaike information criterion score is given by

$$AIC = -2l(\hat{\theta}_n) + 2k.$$

For each distribution, we first compute the MLE of the parameter(s) and then compute the value of maximum likelihood function* and hence the AIC score.

(*In R, one can obtain this value by the following command

logLik(fitdistr(testdata,"distribution name"))

We summarize the test result in the following table:

	MLEs	maximum likelihood function	AIC
(1) Exponential	λ =0.0250156	-4688257	9376516
(2) Normal	$\mu = 39.975113$ $\sigma = 39.98147$	-5107355	10214713
(3) Log-normal	$\mu = 3.110918$ $\sigma = 1.2829$	-4778980	9557964

Since the AIC of exponential distribution has lowest score and we shall choose exponential distribution.

Example 21 (AIC on modelling stock return)

Recall that normal distribution and general-t distribution are two good candidates for modelling stock return.

We consider the logreturn of tencent stock (0700.HK). Using the sample data given, determine the best model using AIC.

😊Solution

Similar to Example 20, we shall compute AIC scores of two models by first obtaining the MLEs of the parameter and compute the AIC score by

$$AIC = -2l(\hat{\theta}_n) + 2k.$$

(Here, $k = 2$ for normal distribution and $k = 3$ for general t distribution)

Here is the sample code for the calculation of AIC score for t-distribution.

```
library("Rfast")
library("ramify")
library("LambertW")
x<-X0700_HK$Logreturn #Store the target data
x<-na.omit(x)           #remove "NA" in the dataset
x<-as.matrix(x)         #ensure that the data is in matrix form
df=seq(2,8,0.001)       #set the range of degree of freedom (e.g.
n=length(df)
```

```
loglik=rep(0,n)          #Used to store the value of loglikelihood function for
different level of df

#For each df, calculate the MLE and loglikelihood function
for(i in 1:n){
  fitresult=mvt.mle(x,v=df[i])
  loglik[i]=fitresult$loglik
}
loglik_value=loglik[which.max(loglik)] #Calculate the maximum loglikelihood value
AIC=-2*loglik_value+2*3 #Calculate the AIC score
print(AIC)

> print(AIC)
[1] -11288.83
```

(*The calculation of AIC score for normal distribution is same as that in Example 20)

We summarized the result in the following table

	MLEs	AIC
Normal Distribution	$\mu = 0.0010897454$ $\sigma = 0.0195335436$	-11134.84
General t Distribution	$\mu = 0.001117649$ $\lambda = 0.01526518$ $\nu = 4.937$	-11288.83

Since AIC score is lower for general t distribution, thus the best model for the logreturn will be general t distribution.

Bayesian Information Criterion (for model selection)

Roughly speaking, this criterion chooses the best model through Bayesian Analysis. It can be seen as an extension from that for parameter estimation.

Problem formulation and the rough derivation of BIC

Suppose that there are N models (denoted by $M_1, M_2, \dots, M_N$) available for a random variable X (with probability density function $f_X(x)$).

For each model M_i, we let $f_X\left(x\middle|\vec{\theta}_i, M_i\right)$ be the associated probability density function of X, where $\vec{\theta}_i = \left(\theta_{i1}, \theta_{i2}, \dots, \theta_{im_i}\right)$ be the parameters.

- We assume that the correct model can be one of these models with equal chance so that the probability that the correct model is M_i is $P(M = M_i) = \frac{1}{N}$ (prior distribution)
- For each model M_i, we let $\pi\left(\vec{\theta}_i\right)$ be the prior distribution of $\vec{\theta}_i$. We assume that the prior distribution of $\vec{\theta}_i$ is flat so that $\pi\left(\vec{\theta}_i\right) = c_i$, where $c_i > 0$ is some constant.

Given the sample data $\{X_1, X_2, \dots, X_n\}$, one can determine the posterior distribution of the model M:

$$P(M = M_i | X_1, X_2, \dots, X_n) = \frac{f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i) \overbrace{P(M = M_i)}^{=1/N}}{f_{\vec{X}}(X_1, X_2, \dots, X_n)} \dots\dots (*)$$

Here, $f_{\vec{X}}(x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} f_X(x_i)$ denotes the joint density function of $X_1, \dots, X_n$.

Intuitively, one should choose the model M_i with the highest posterior probability.

Since the denominator in $(*)$ is independent of M_i, so it suffices to choose M_i such that the function $f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i)$ is maximized.

Formula for estimating $f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i)$ (given M_i)

Recall that the prior distribution of $\vec{\theta}_i$ is $\pi(\vec{\theta}_i) = c_i$, then one can deduce that

$$f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i) = \int \underbrace{f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i, \vec{\theta}_i)}_{Likelihood\ function\ L(\vec{\theta}_i)} \underbrace{\pi(\vec{\theta}_i)}_{=c_i} d\vec{\theta}_i \dots\dots (**)$$

(*Here, the above integral should be understood as multiple integral with respect to $\vec{\theta}_i$ over all possible range of $\vec{\theta}_i$)

To proceed, we let $l(\vec{\theta}_i) = \ln L(\vec{\theta}_i)$ be the loglikelihood function and let $\vec{\theta}_i^* = \vec{\theta}_i^*(X_1, X_2, \dots, X_n)$ be the MLE of $\vec{\theta}_i$.

Recall that $\vec{\theta}_i^*$ maximizes the loglikelihood function such that $\frac{\partial l}{\partial \theta_{ij}}|_{\vec{\theta}_i = \vec{\theta}_i^*} = 0$ for $j = 1,2, \dots, m_i$. One can consider the Taylor explanation of $l(\vec{\theta}_i)$ at $\vec{\theta}_i^*$ and obtain

$$l(\vec{\theta}_i) = l(\vec{\theta}_i^*) + \sum_{j=1}^{m_i} \underbrace{\frac{\partial l}{\partial \theta_{ij}}|_{\vec{\theta}_i = \vec{\theta}_i^*}}_{=0} (\theta_{ij} - \theta_{ij}^*) + \frac{1}{2}\sum_{j=1}^{m_i}\sum_{k=1}^{m_i} \frac{\partial^2 l}{\partial \theta_{ij}^2}|_{\vec{\theta}_i = \vec{\theta}_i^*} (\theta_{ij} - \theta_{ij}^*)(\theta_{ik} - \theta_{ik}^*)$$

$$+ \cdots\cdots.$$

$$\Rightarrow l(\vec{\theta}_i) \approx l(\vec{\theta}_i^*) + \frac{1}{2}(\vec{\theta}_i - \vec{\theta}_i^*)^T H(\vec{\theta}_i^*)(\vec{\theta}_i - \vec{\theta}_i^*)$$

Here, $H(\vec{\theta}_i^*) = \begin{pmatrix} \frac{\partial^2 l}{\partial \theta_{i1}^2} & \frac{\partial^2 l}{\partial \theta_{i1}\partial \theta_{i2}} & \cdots & \frac{\partial^2 l}{\partial \theta_{i1}\partial \theta_{im_i}} \\ \frac{\partial^2 l}{\partial \theta_{i2}\partial \theta_{i1}} & \frac{\partial^2 l}{\partial \theta_{i2}^2} & \cdots & \frac{\partial^2 l}{\partial \theta_{i2}\partial \theta_{im_i}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 l}{\partial \theta_{im_1}\partial \theta_{i1}} & \frac{\partial^2 l}{\partial \theta_{im_1}\partial \theta_{i2}} & \cdots & \frac{\partial^2 l}{\partial \theta_{im_i}^2} \end{pmatrix}$ denotes the Hessian matrix of l.

Since $l(\vec{\theta}_i)$ is maximized at $\vec{\theta}_i = \vec{\theta}_i^*$, so $H(\vec{\theta}_i^*)$ is negatively definite (second-order condition). To facilitate the later calculation, we let $\widetilde{H}(\vec{\theta}_i^*) = -H(\vec{\theta}_i^*)$. Then $\widetilde{H}(\vec{\theta}_i^*)$ is positively definite. Substituting the result into $(**)$, we have

$$f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i) = c_i \int L(\vec{\theta}_i) d\vec{\theta}_i = c_i \int e^{\overbrace{\ln L(\vec{\theta}_i)}^{l(\vec{\theta}_i)}} d\vec{\theta}_i$$

$$= c_i \int e^{l(\vec{\theta}_i^*) + \frac{1}{2}(\vec{\theta}_i - \vec{\theta}_i^*)^T H(\vec{\theta}_i^*)(\vec{\theta}_i - \vec{\theta}_i^*)} d\vec{\theta}_i = c_i \int e^{l(\vec{\theta}_i^*) - \frac{1}{2}(\vec{\theta}_i - \vec{\theta}_i^*)^T \widetilde{H}(\vec{\theta}_i^*)(\vec{\theta}_i - \vec{\theta}_i^*)} d\vec{\theta}_i \dots (***)$$

To move on, we note that $\frac{\partial^2 l}{\partial \theta_{ij} \partial \theta_{ik}} = \frac{\partial^2 l}{\partial \theta_{ik} \partial \theta_{ij}}$ (assuming $l(\vec{\theta}_i)$ is sufficiently smooth), then the matrix $\widetilde{H}$ (or H) is symmetric. It follows that $\widetilde{H}$ can be diagonalized into

$$\widetilde{H}(\vec{\theta}_i^*) = UDU^T,$$

where $D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_{m_i} \end{pmatrix}$, $\lambda_i > 0$ are eigenvalues of $\widetilde{H}$ and U is orthogonal matrix which the columns are eigenvectors of $\widetilde{H}$.

Using change of variable and the direct integration, we deduce that

$$f_{\vec{X}}(X_1, X_2, \dots, X_n | M_i) = c_i \int e^{l(\vec{\theta}_i^*) - \frac{1}{2}(\vec{\theta}_i - \vec{\theta}_i^*)^T \tilde{H}(\vec{\theta}_i^*)(\vec{\theta}_i - \vec{\theta}_i^*)} d\vec{\theta}_i$$

$$= c_i e^{l(\vec{\theta}_i^*)} \int e^{-\frac{1}{2}(\vec{\theta}_i - \vec{\theta}_i^*)^T \tilde{H}(\vec{\theta}_i^*)(\vec{\theta}_i - \vec{\theta}_i^*)} d\vec{\theta}_i = c_i e^{l(\vec{\theta}_i^*)} \int e^{-\frac{1}{2}(\vec{\theta}_i - \vec{\theta}_i^*)^T UDU^T (\vec{\theta}_i - \vec{\theta}_i^*)} d\vec{\theta}_i$$

$$\overset{\vec{y} = U^T(\vec{\theta}_i - \vec{\theta}_i^*)}{=\!=\!=} c_i e^{l(\vec{\theta}_i^*)} \int e^{-\frac{1}{2}\vec{y}^T D \vec{y}} \underbrace{|\det U|}_{=1^{(*)}} d\vec{y} = c_i e^{l(\vec{\theta}_i^*)} \int e^{-\frac{1}{2}\left(\lambda_1 y_1^2 + \cdots + \lambda_{m_i} y_{m_i}^2\right)} d\vec{y}$$

$$= \cdots =^{(?)} c_i e^{l(\vec{\theta}_i^*)} \prod_{k=1}^{m_i} \sqrt{\frac{2\pi}{\lambda_k}}$$

$$= c_i e^{l(\vec{\theta}_i^*)} \frac{(2\pi)^{\frac{m_i}{2}}}{\prod_{k=1}^{m_i} \sqrt{\lambda_k}}$$

$$= c_i L(\vec{\theta}_i^*) \frac{(2\pi)^{\frac{m_i}{2}}}{(\det \tilde{H})^{\frac{1}{2}}} \dots\dots (****)$$

We used the fact that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = 1 \Rightarrow \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz = \sqrt{2\pi}$$

$$\det \tilde{H} = (\det U)(\det D)(\det U^T) = \det(D) = \lambda_1 \lambda_2 \dots \lambda_{m_i}$$

(*Note: Since U is orthogonal matrix with $UU^T = I$, it follows that $\det U = \pm 1$)

Finally, we will simplify the formula $(****)$ and obtain a more user friendly formula for estimating $f_{\vec{X}}(X_1, X_2, \dots, X_n|M_i)$.

To start with, we consider

$$\ln f_{\vec{X}}(X_1, X_2, \dots, X_n|M_i) = \ln c_i + \underbrace{\ln L\left(\vec{\theta}_i^*\right)}_{=l\left(\vec{\theta}_i^*\right)} + \frac{m_i}{2}\ln 2\pi - \frac{1}{2}\ln\det\widetilde{H}.$$

We consider the second term $l\left(\vec{\theta}_i^*\right) = \sum_{k=1}^{n}\ln f_X\left(X_k\middle|M_i, \vec{\theta}_i^*\right)$.

- Assume that the sample size n is very large, we use strong law of large number and get

$$l\left(\vec{\theta}_i\right) = \sum_{k=1}^{n}\ln f_X\left(X_k\middle|M_i, \vec{\theta}_i\right) = n\left(\frac{\sum_{k=1}^{n}\ln f_X\left(X_k\middle|M_i, \vec{\theta}_i\right)}{n}\right) \approx n\mathbb{E}\left[\ln f_X\left(X\middle|M_i, \vec{\theta}_i\right)\right].$$

Next, we consider the last term $\ln\det\widetilde{H}$.

- Using law of large number again, we have

$$-\frac{\partial^2 l}{\partial\theta_{ij}\partial\theta_{ik}} = -\frac{\partial^2}{\partial\theta_{ij}\partial\theta_{ik}}\left(\sum_{k=1}^{n}\ln f_X\left(X_k\middle|M_i, \vec{\theta}_i\right)\right) = -n\left(\frac{\sum_{k=1}^{n}\frac{\partial^2}{\partial\theta_{ij}\partial\theta_{ik}}\ln f_X\left(X_k\middle|M_i, \vec{\theta}_i\right)}{n}\right)$$

$$\approx -n\mathbb{E}\left[\frac{\partial^2}{\partial\theta_{ij}\partial\theta_{ik}}\ln f_X\left(X\middle|M_i, \vec{\theta}_i\right)\right].$$

- Then the determinant of matrix $\widetilde{H}$ can be expressed as

$$\det \widetilde{H} = \det \begin{pmatrix} -\frac{\partial^2 l}{\partial \theta_{i1}^2} & \cdots & -\frac{\partial^2 l}{\partial \theta_{i1}\partial \theta_{im_i}} \\ \vdots & \ddots & \vdots \\ -\frac{\partial^2 l}{\partial \theta_{im_i}\partial \theta_{i1}} & \cdots & -\frac{\partial^2 l}{\partial \theta_{im_i}^2} \end{pmatrix}$$

$$= \det \begin{pmatrix} -n\mathbb{E}\left[\frac{\partial^2}{\partial \theta_{i1}^2} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] & \cdots & -n\mathbb{E}\left[\frac{\partial^2 l}{\partial \theta_{i1}\partial \theta_{im_i}} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] \\ \vdots & \ddots & \vdots \\ -n\mathbb{E}\left[\frac{\partial^2 l}{\partial \theta_{im_i}\partial \theta_{i1}} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] & \cdots & -n\mathbb{E}\left[\frac{\partial^2}{\partial \theta_{im_i}^2} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] \end{pmatrix}$$

$$= n^{m_i} \underbrace{\det \begin{pmatrix} -\mathbb{E}\left[\frac{\partial^2}{\partial \theta_{i1}^2} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] & \cdots & -\mathbb{E}\left[\frac{\partial^2 l}{\partial \theta_{i1}\partial \theta_{im_i}} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] \\ \vdots & \ddots & \vdots \\ -\mathbb{E}\left[\frac{\partial^2 l}{\partial \theta_{im_i}\partial \theta_{i1}} \ln f_X\left(X \middle| M_i, \vec{\theta}_i\right)\right] & \cdots & -\mathbb{E}\left[\frac{\partial^2}{\partial \theta_{im_i}^2} \ln f_X\left(X \middle| M_i, \vec{\theta}_{\square\square}\right)\right] \end{pmatrix}}_{Denote\ it\ by\ \det A\ \ (just\ a\ fixed\ number!)}$$

- Thus, we have

$$\ln \det \widetilde{H} = m_i \ln n + \ln \det A.$$

Combining these approximation, we get

$$\ln f_{\vec{X}}(X_1, X_2, \dots, X_n|M_i) = \ln c_i + n\mathbb{E}\left[\ln f_X\left(X\middle|M_i, \vec{\theta}_i^*\right)\right] + \frac{m_i}{2}\ln 2\pi - \frac{1}{2}(m_i \ln n + \ln\det A).$$

Since n is assumed to be very large, so we can omit the terms which are independent to n. Then the formula can be simplified into

$$\ln f_{\vec{X}}(X_1, X_2, \dots, X_n|M_i) = n\mathbb{E}\left[\ln f_X\left(X\middle|M_i, \vec{\theta}_i^*\right)\right] - \frac{m_i}{2}\ln n \approx l\left(\vec{\theta}_i^*\right) - \frac{m_i}{2}\ln n$$
$$= -\frac{1}{2}\left(m_i \ln n - 2l\left(\vec{\theta}_i^*\right)\right)$$

We define BIC score to be

$$BIC = -2\begin{pmatrix} maximum \\ loglikelihood \end{pmatrix} + m_i \ln n.$$

Then the best model will be the one with the lowest BIC score (so that $f_{\vec{X}}(X_1, X_2, \dots, X_n|M_i)$ is maximized)

- As a side note, we observe that BIC score is very similar to AIC score except that the "penalty term" (second term) is $m_i \ln n$ instead of $2m_i$.

105 MATH4515 Statistical and Computational Methods in Financial Mathematics

Topic 1: Basic Statistcal Methods for Model Construction